MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
4000 PROCESS AREA: ADMINISTRATION BUILDING
DESIGN OF BUILDING STRUCTURE

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TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 4
1. INTENT 5
2. EXECUTIVE SUMMARY 5
3. SCOPE 5
4. DESIGN METHODOLOGY & ASSUMPTIONS 6
4.1. Codes 6
4.2. Geotechnical investigations 7
4.3. Material Properties 7
5. DESIGN LOADS 7
5.1. Dead Load(DL) 7
5.1.1. Roof Dead Loads 7
5.1.2. Ground Floor Walls 10
5.2. Live Load 10
5.2.1. Roof Live Loads(Lr) 10
5.3. Wind Load(WL) 11
5.4. Wind Load (Analytical Verification) 11
5.5. Seismic Load(E) 15
5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads) 15
6. DESIGN COMBINATIONS 16
7. CRACKED SECTIONS 17
8. DEFLECTIONS 17
8.1. Maximum Allowable Deflection 17
9. CONCRETE ROOF BEAM DESIGN 18
9.1. Design of Ledger Beam(LB1) 18
9.1.1. Staad Output For LB1 20
9.2. Design of Ledger Beam(LB2 & LB3) 21
9.2.1. Long Term Deflection Check For LB2/LB3 23

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9.2.2. Staad Output For LB2....27
9.2.3. Staad Output For LB3....28
9.3. Design of Ledger Beam(LB4)....29
9.3.1. Staad Output For LB4....30
9.4. Design of Beam-(B1)....32
9.4.1. Staad Output For B1....33
9.5. Design of Beam-(B2)....34
9.5.1. Staad Output For B2....35
10. CONCRETE GRADE SLAB DESIGN....36
11. DESIGN OF GRADE BEAM....37
11.1. Grade Beam(GB1)....37
11.1.1. STAAD OUTPUT FOR GB1....38
11.2. Grade Beam(GB2)....39
11.2.1. STAAD OUTPUT FOR GB1....40
12. CONCRETE COLUMN DESIGN....41
12.1. Staad Output For Column C1 & CR1....43
12.2. Staad Output For Column C2 & CR2....45
12.3. Staad Output For Column CR3....45
13. CANTILEVER SLAB DESIGN....46
14. FOUNDATION DESIGN....48
References....49
ATTACHMENT 1 STAAD INPUT DATA....50
ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS....58

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Administration Building.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Administration Building only.

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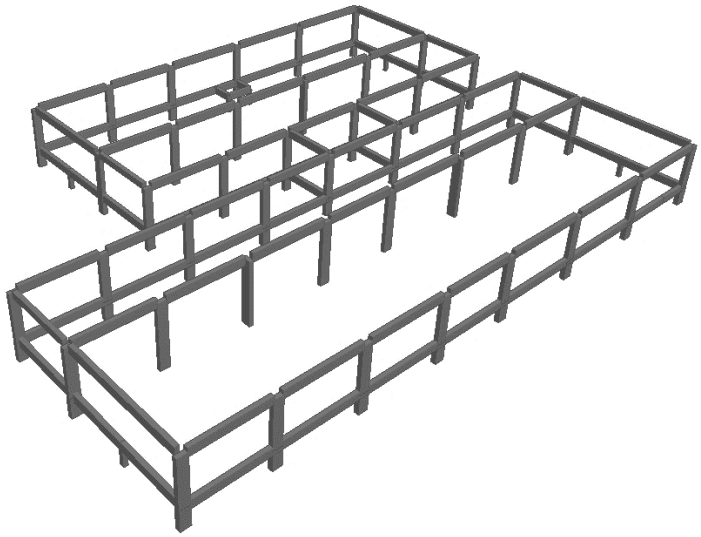
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4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Building was modelled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-A02.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

5. DESIGN LOADS

5.1. Dead Load(DL)

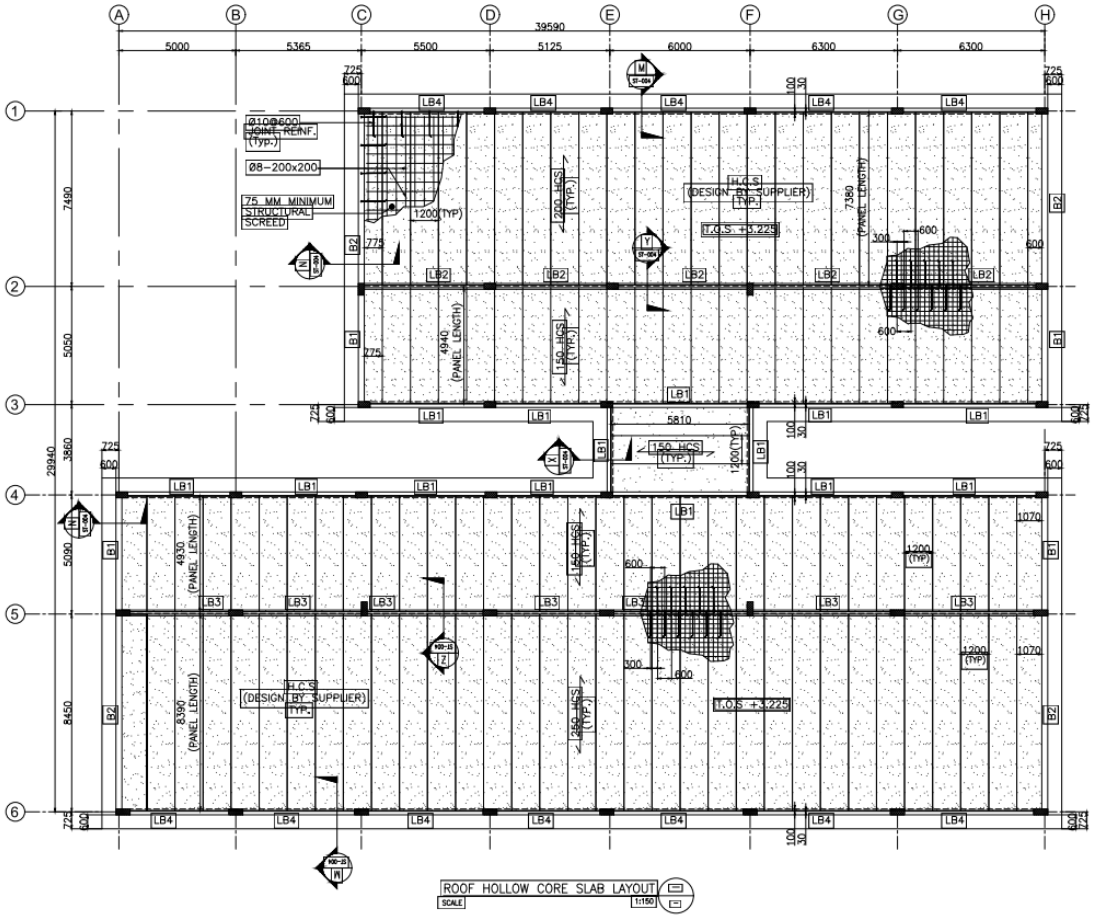
5.1.1. Roof Dead Loads

• Provision for Electrical Lighting	= 0.05 kN/m²
Ceiling	
Suspended steel channel system	= 0.10 kN/m²
Acoustical Fiber Tile (16mm)	= 0.05 kN/m²
Sub-total	= **0.2 kN/m²**

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- Hollow Core Slab

  (HCS-200)(To be confirmed by Vendor) b/w Grid Line's (1 & 2) = **3.10 KN/m2**

  (HCS-150)(To be confirmed by Vendor) b/w Grid Line's (2 & 3), (4 & 5) = **2.52 KN/m2**

  (HCS-150)(To be confirmed by Vendor) b/w Grid Line's (E & F) = **2.52 KN/m2**

  (HCS-250)(To be confirmed by Vendor) b/w Grid Line's (5 & 6) = **3.43 KN/m2**

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Prestressed Hollowcore Floor

Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/ m²**

Therefore

Total Dead Load b/w Grid's (1 & 2) =3.1+0.2+3.1= 6.40 **kN/m²**

Total Dead Load b/w Grid's (2 & 3), (4 & 5) =2.52+0.2+3.1= 5.82 **kN/m²**

Total Dead Load b/w Grid's (E & F) =2.52+0.2+3.1= 5.82 **kN/m²**

Total Dead Load b/w Grid's (5 & 6) =3.43+0.2+3.1= 6.73 **kN/m²**

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TOTAL DL as UDL ON BEAMS

- UDL on Ledger Beams Along Grid Line-1 = 6.40 x 7.50/2 = **24 Kn/m**
- UDL on Ledger Beams Along Grid Line-6 = 6.73 x 8.45/2 =**28.43 Kn/m**
- UDL on Ledger Beams Along Grid Line's-3,4 =5.82 x 5.10/2 =**14.9 Kn/m**
- UDL on Ledger Beams Along Grid Line 5=28.4+14.9 =**43.32 Kn/m**
- UDL on Ledger Beams Along Grid Line 2=24+14.9 =**38.9 Kn/m**
- UDL on Ledger Beams Along Grid Line's E,F=5.82 x 6.00/2 =**19.2 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

5.1.2. Ground Floor Walls

- 250 tyke. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x (3.5-0.5) m clear height = **12 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads(Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

TOTAL Lr as UDL ON BEAMS

- UDL on Ledger Beams Along Grid Line-1 = 0.50 x 7.50/2 = **1.875 Kn/m**
- UDL on Ledger Beams Along Grid Line-6 = 0.50 x 8.45/2 =**2.11 Kn/m**
- UDL on Ledger Beams Along Grid Line's-3,4 =0.50 x 5.10/2 =**1.275 Kn/m**
- UDL on Ledger Beams Along Grid Line 5=1.275+2.11 =**3.385 Kn/m**
- UDL on Ledger Beams Along Grid Line 2=1.875+1.275 =**3.15 Kn/m**
- UDL on Ledger Beams Along Grid Line's E,F=0.50 x 6.00/2 =**1.50 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

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5.3. Wind Load(WL)

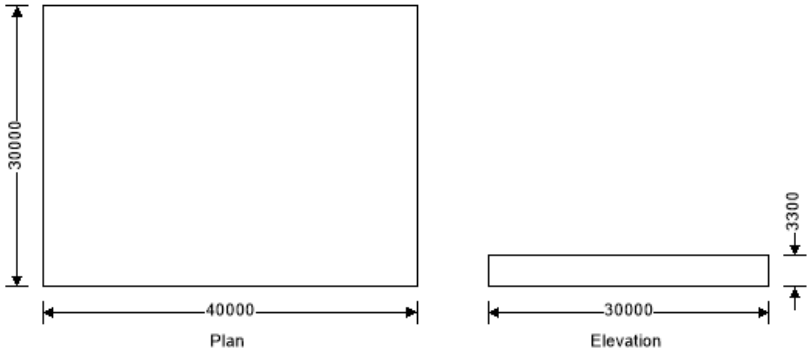
Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 40000 mm
Width of building;	d = 30000 mm
Height to eaves;	H = **3300** mm
Mean height;	h = **3300** mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)

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Exposure category; D(As per SBC-301, Table 6.4.2.3)

Enclosure classification; Partially enclosed buildings(As per SBC-301, Section-6.2)

Internal pressure coef +ve; GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)

Internal pressure coef –ve; GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)

Gust effect factor; G_f = **0.85**(As per SBC-301, Section-7.2.7.1)

Importance Factor; 1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3300	1.03	1.29
3301	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	132.03	22.12
B	3.30	-0.50	1.29	-1.26	132.00	-165.84
C	3.30	-0.70	1.29	-1.48	99	-146.07
D	3.30	-0.70	1.29	-1.48	99	-146.07

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Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	132.04	209.27
B	3.30	-0.50	1.29	0.16	132	21.26
C	3.30	-0.70	1.29	-0.06	99	-5.74
D	3.30	-0.70	1.29	-0.06	99	-5.74

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	99.03	16.59
B	3.30	-0.21	1.29	-1.18	99.00	-117.15
C	3.30	-0.70	1.29	-1.48	132.00	-194.75
D	3.30	-0.70	1.29	-1.48	132.00	-194.75

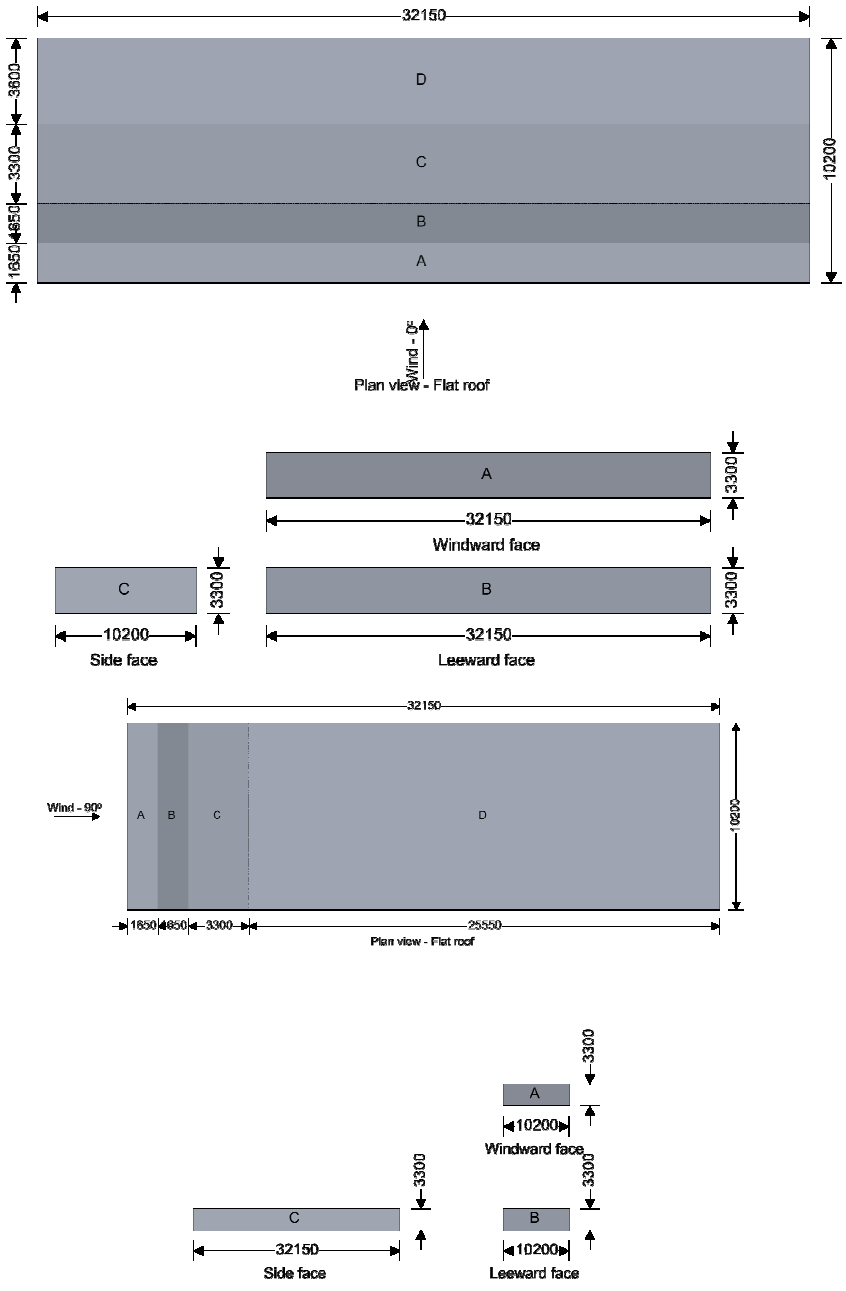
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	99.03	156.96
B	3.30	-0.43	1.29	0.23	99.00	23.17
C	3.30	-0.70	1.29	-0.06	132.00	-7.65
D	3.30	-0.70	1.29	-0.06	132.00	-7.65

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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.5. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

8. DEFLECTIONS

8.1. Maximum Allowable Deflection

L/240 : for member load with live load plus dead load

L/600 : for members supporting masonry under total loads

Where, L = span of member between support

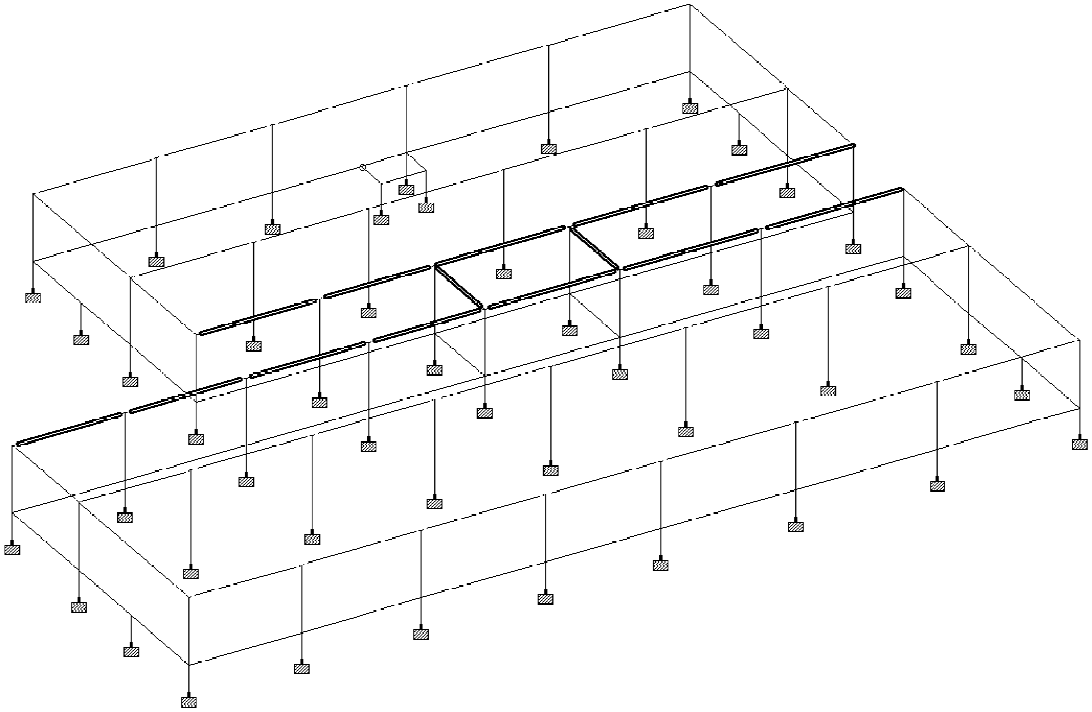
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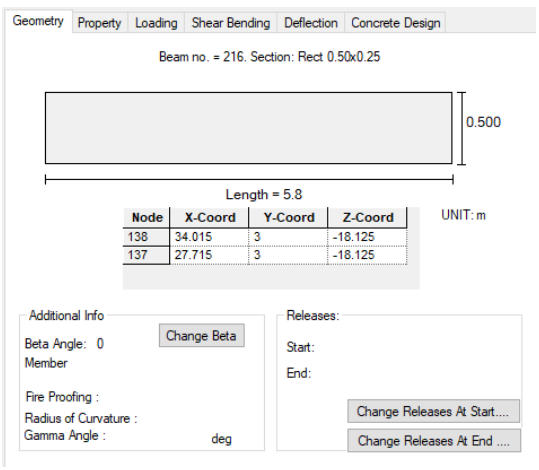
9. CONCRETE ROOF BEAM DESIGN

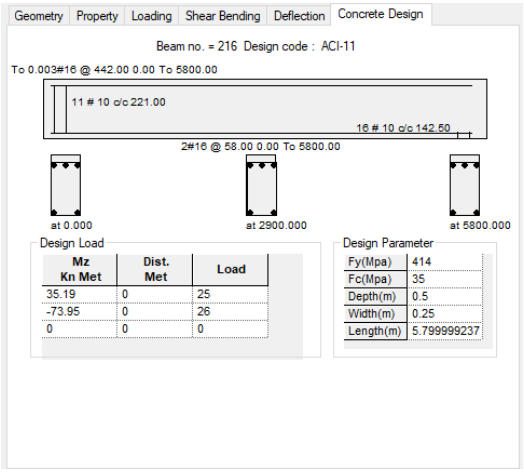
9.1. Design of Ledger Beam(LB1)



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Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 73 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **476 mm^2**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm^2**

Maximum Moment required from Analysis = **35 Knm (Sagging)**

Ast2.req = **238 mm^2**

Hence Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bottom

Ast1.pro= **602mm^2 >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **602mm^2 >>>>>>(Ast.min, Ast2.req)**

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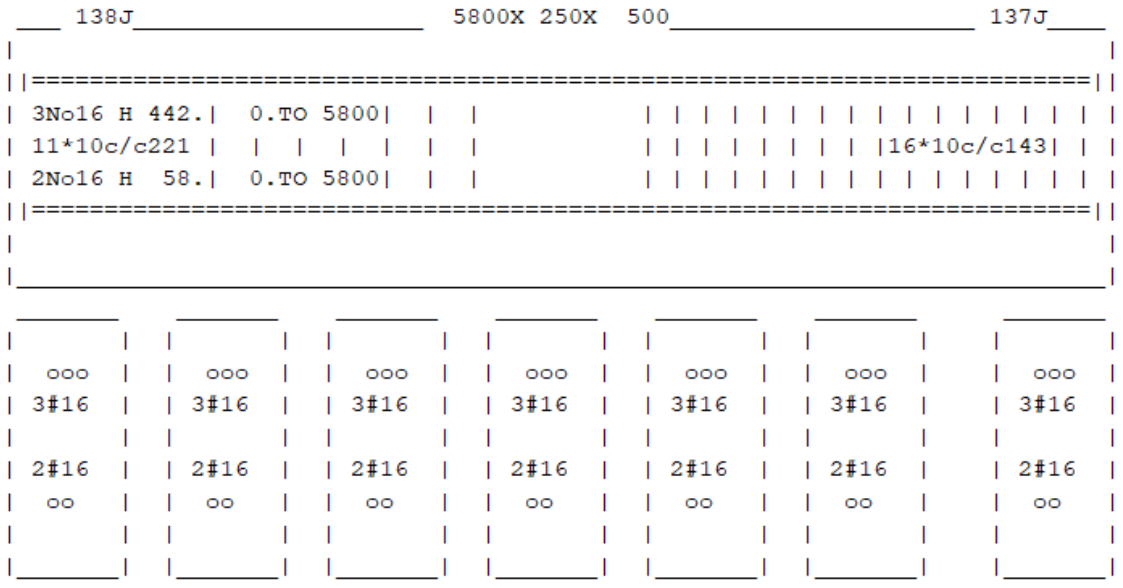
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9.1.1. Staad Output For LB1

```
            B E A M   N O.   216 D E S I G N   R E S U L T S - SHEAR


  AT START SUPPORT - Vu=    50.93 KNS   Vc=  111.93 KNS   Vs=     0.00 KNS
  Tu=     1.84 KN-MET   Tc=     4.0 KN-MET   Ts=     0.0 KN-MET   LOAD     24
  NO STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2020. MM

  AT END    SUPPORT - Vu=    42.87 KNS   Vc=  112.98 KNS   Vs=     0.00 KNS
  Tu=     6.53 KN-MET   Tc=     4.0 KN-MET   Ts=     8.7 KN-MET   LOAD     32
  STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2020. MM
  ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    2.15 SQ.CM.
```

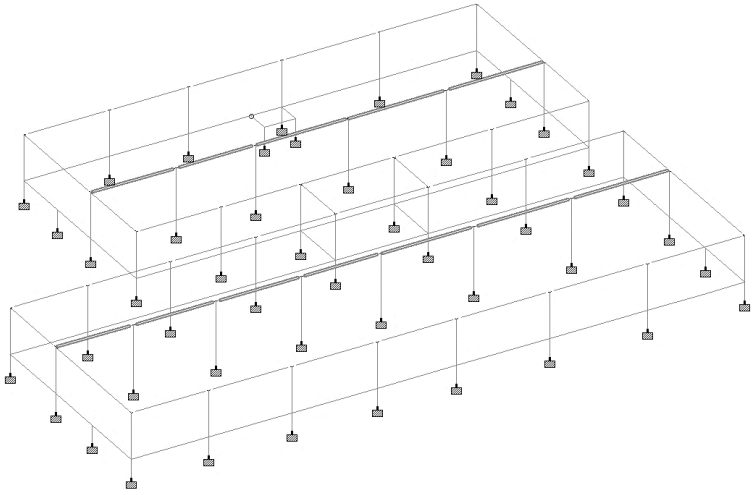


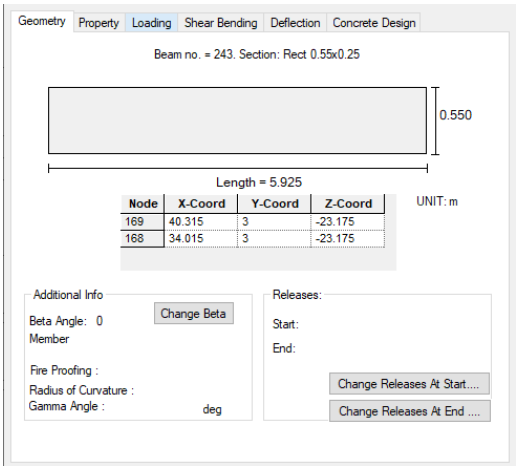
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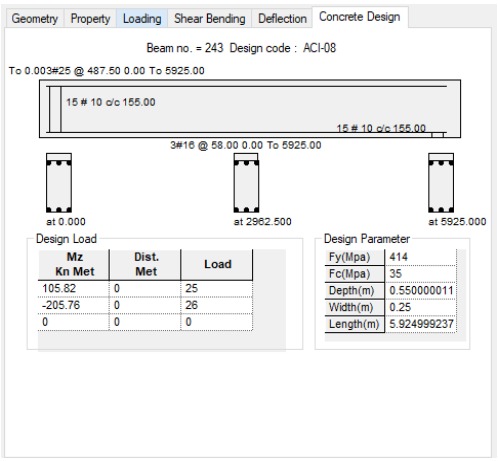
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9.2. Design of Ledger Beam(LB2 & LB3)







Design of Ledger Beam (LB2/LB3)

Maximum Moment required from Analysis = 205 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=414 Mpa, d=550-40-10-20/2=490mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **1205 mm^2**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **437mm^2**

Maximum Moment required from Analysis = **105 Knm (Sagging)**

Ast2.req = **595 mm^2**

However Provide 3-25 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom, 2-20 dia-Addit Bottom..for Beams LB2 & LB3 considering Long term Deflection check in Section 8.2.1

Ast1.pro**>>>>>>(Ast.min, Ast1.req)**

Ast2.pro **>>>>>>(Ast.min, Ast2.req)**

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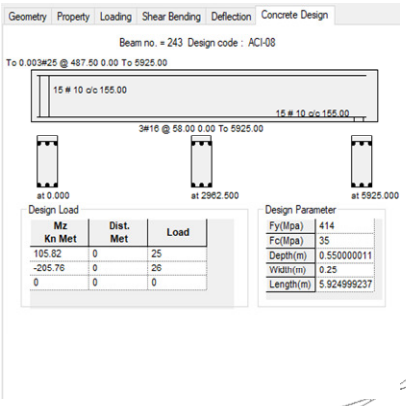
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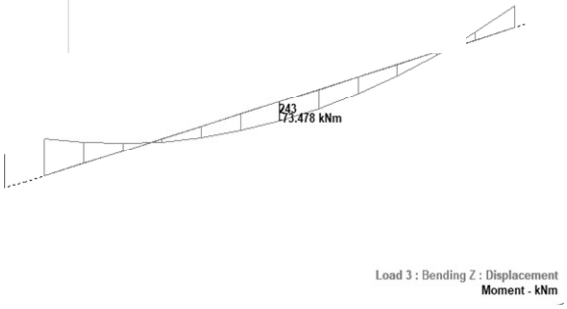
9.2.1. Long Term Deflection Check For LB2/LB3

(LB2-BEAM-LONGTERM DEFLECTION CHECK)

INPUT DATA :

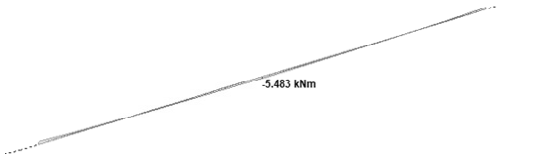
Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	550	mm
width of beam, b	=	250	mm
Length of beam, L	=	6.00	m
Bottom bar diam., db	=	16.00	mm
area of 1 bar Ab	=	201.06	mm^2
no. of bottom bars	=	5.00	pcs.
Top bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	492.00	mm
(FROM STAAD RESULTS) Mdead	=	75.00	kN-m
Mlive	=	6.00	kN-m





CHECK DEFLECTION: $M_{d+\lambda}$ = 81.00 kN-m

$M_{sus} = M_d + 0.50M_\lambda$ = 78.00 kN-m



modulus of rupture, modulus of elasticity, modular ratio

$fr = 0.62\lambda\sqrt{fc'}$ = 3.67 Mpa



$n_s = \frac{Es}{Ec}$ = 7.19 ←*ACI(318M-08) Sec. 9.5.2.3*

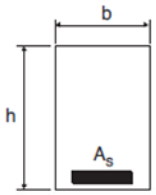
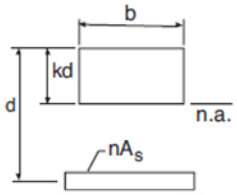
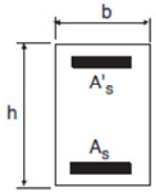
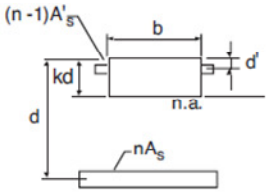
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 3.47E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.03 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 142.24 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 1.12E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 3.08$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 46.23 \text{ kN-m}$$ ←*ACI(318M-08) Sec. 9.5.2.3*

under dead load

$$\frac{M_{cr}}{M_d} = 0.62$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 1.67E+09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.59$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 1.61E+09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

under dead + live load

$$\frac{M_{cr}}{M_{d+\lambda}} = 0.57$$

$$(I_e)_{d+\lambda} = \left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3\right] I_{cr} = 1.56E+09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\lambda^2}{Ec(I_e)_d} = 6.046 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\lambda^2}{Ec(I_e)_{sus}} = 6.526 \text{ mm}$$

$$(\Delta_i)_{d+\lambda} = \frac{K(5/48)M_{d+\lambda}\lambda^2}{Ec(I_e)_{d+\lambda}} = 7.003 \text{ mm}$$

$$(\Delta_i)_\lambda = (\Delta_i)_{d+\lambda} - (\Delta_i)_d = 0.957 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$$(\Delta_i)_\lambda \le \frac{\lambda}{180} = 33.33 \text{ mm} > 0.96 \text{ mm} \quad ok!$$

$$(\Delta_i)_\lambda \le \frac{\lambda}{360} = 16.67 \text{ mm} > 0.96 \text{ mm} \quad ok!$$

additional long-term deflection

$\xi = 2.00$

5 years or more 2.0
12 months..... 1.4
6 months..... 1.2
3 months..... 1.0

$$\rho = \frac{As}{bd} = 0.0082$$

$$\lambda_\Delta = \frac{\xi}{1+50\rho'} = 1.42$$

←*ACI(318M-08) Sec. 9.5.2.5*

$$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus} = 9.27 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda = 10.22 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{480} = 12.50 \text{ mm} > 10.22 \text{ mm} \quad ok!$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{240} = 25.00 \text{ mm} > 10.22 \text{ mm} \quad ok!$$

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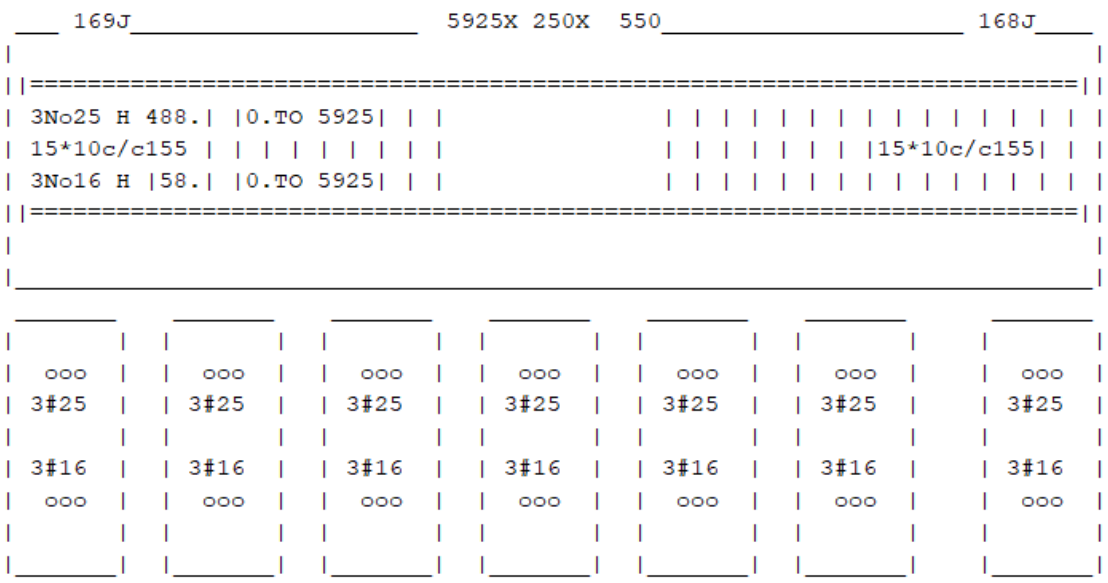
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9.2.2. Staad Output For LB2

```
        B E A M   N O.   243 D E S I G N   R E S U L T S - SHEAR


AT START SUPPORT - Vu=   89.93 KNS  Vc=  218.98 KNS  Vs=    0.00 KNS
Tu=   14.60 KN-MET  Tc=    4.6 KN-MET  Ts=   19.5 KN-MET  LOAD    32
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 155. MM  C/C FOR 2032. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.66 SQ.CM.

AT END   SUPPORT - Vu=  115.63 KNS  Vc=  131.29 KNS  Vs=   22.88 KNS
Tu=   14.60 KN-MET  Tc=    4.6 KN-MET  Ts=   19.5 KN-MET  LOAD    32
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 155. MM  C/C FOR 2032. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.66 SQ.CM.
```



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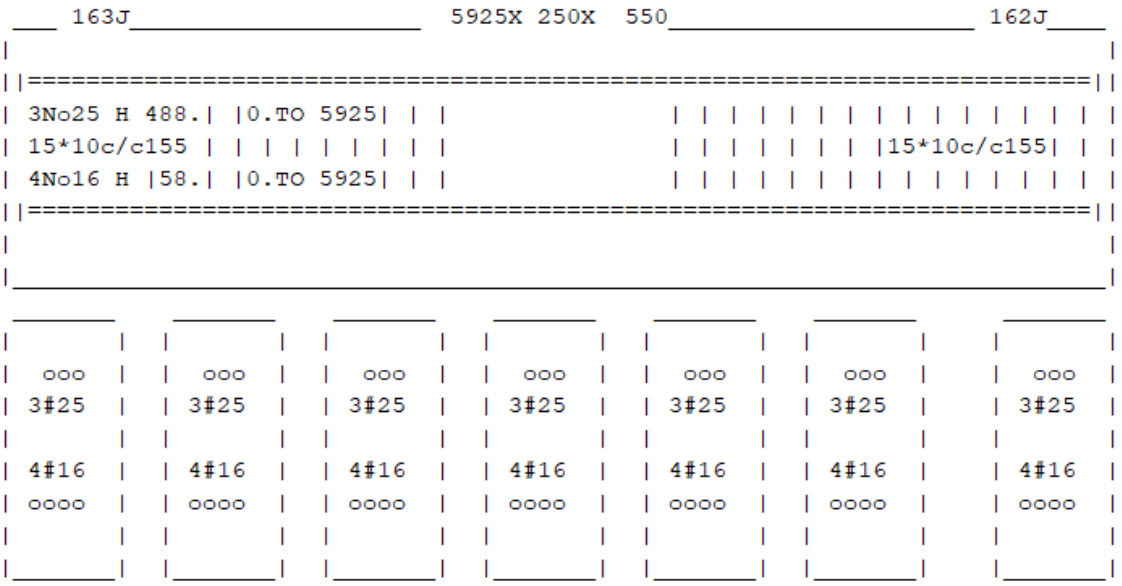
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9.2.3. Staad Output For LB3

```
            B E A M   N O.   238 D E S I G N   R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   78.09 KNS  Vc=  219.20 KNS  Vs=    0.00 KNS
 Tu=   10.01 KN-MET  Tc=    4.6 KN-MET  Ts=   13.4 KN-MET  LOAD    35
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 155. MM  C/C FOR 2032. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   3.20 SQ.CM.

 AT END   SUPPORT - Vu=  123.00 KNS  Vc=  134.97 KNS  Vs=   29.04 KNS
 Tu=    9.69 KN-MET  Tc=    4.7 KN-MET  Ts=   12.9 KN-MET  LOAD    32
 STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 155. MM  C/C FOR 2032. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   3.10 SQ.CM.
```

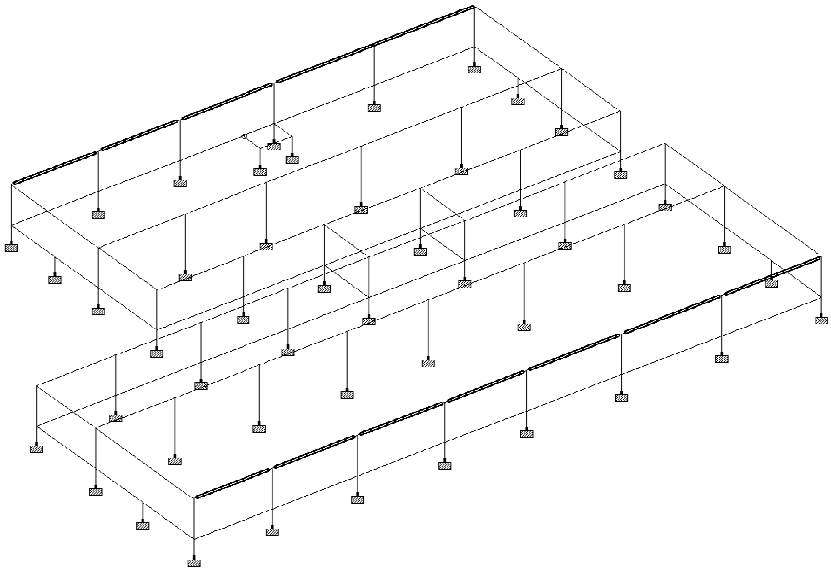


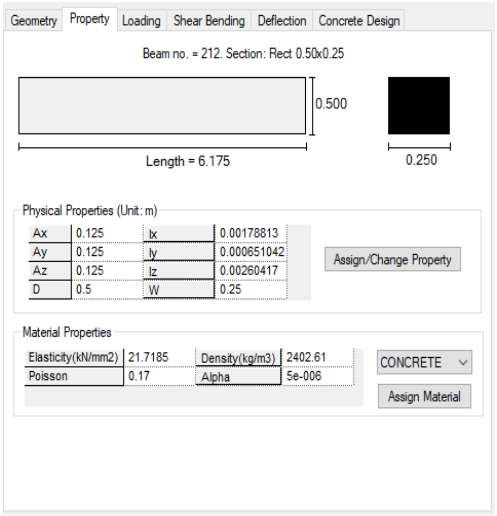
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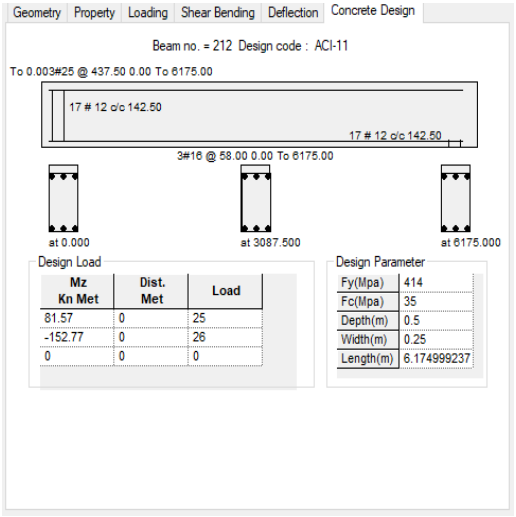
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9.3. Design of Ledger Beam(LB4)







Design of Ledger Beam (LB4)

Maximum Moment required from Analysis = 152 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **990 mm^2**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm^2**

Maximum Moment required from Analysis = **81 Knm (Sagging)**

Ast2.req = **540 mm^2**

Hence Provide 3-25 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast1.pro= **1470mm^2 >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **942mm^2 >>>>>>(Ast.min, Ast2.req)**

9.3.1. Staad Output For LB4

```
        B E A M   N O.   212 D E S I G N   R E S U L T S - SHEAR

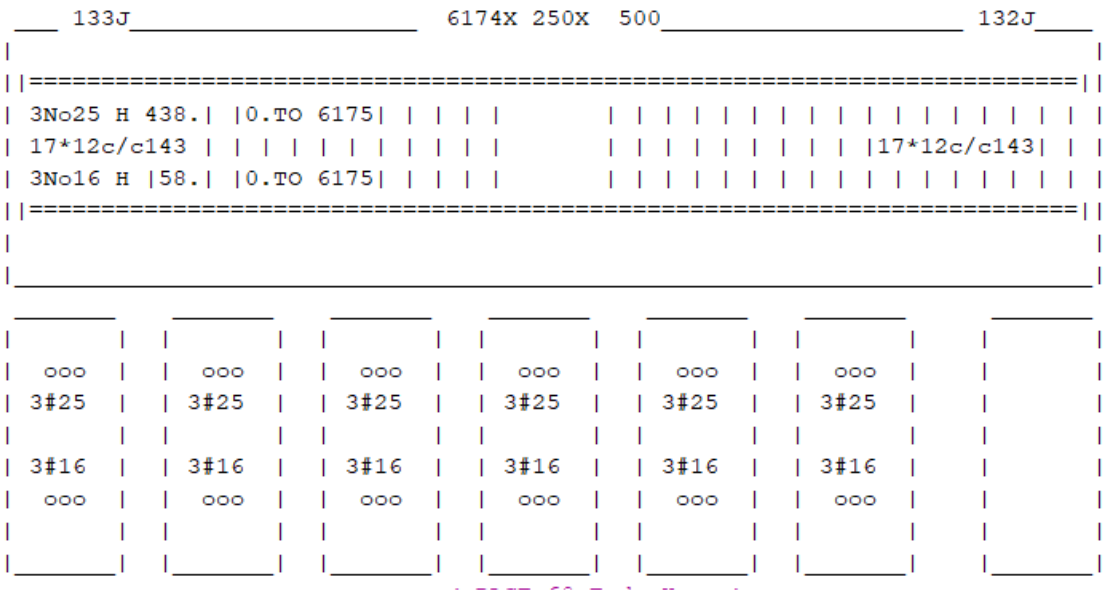

AT START SUPPORT - Vu=    59.40 KNS   Vc=  131.20 KNS   Vs=     0.00 KNS
Tu=    23.30 KN-MET   Tc=     4.2 KN-MET   Ts=    31.1 KN-MET   LOAD     49
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2208. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    7.68 SQ.CM.

AT END    SUPPORT - Vu=    87.50 KNS   Vc=  112.36 KNS   Vs=     4.30 KNS
Tu=    17.78 KN-MET   Tc=     4.2 KN-MET   Ts=    23.7 KN-MET   LOAD     49
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2208. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    5.86 SQ.CM.
```


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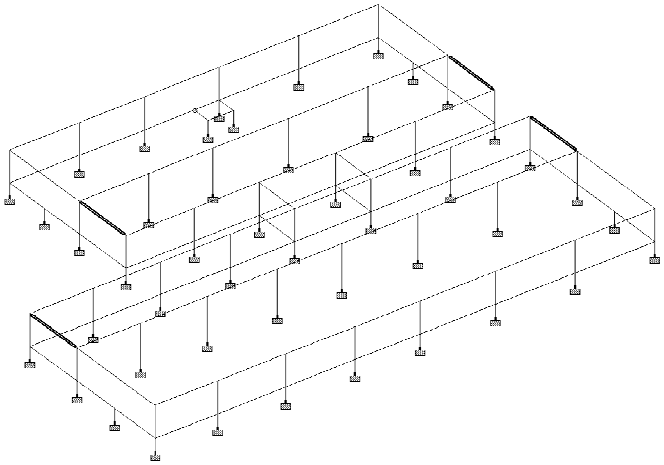


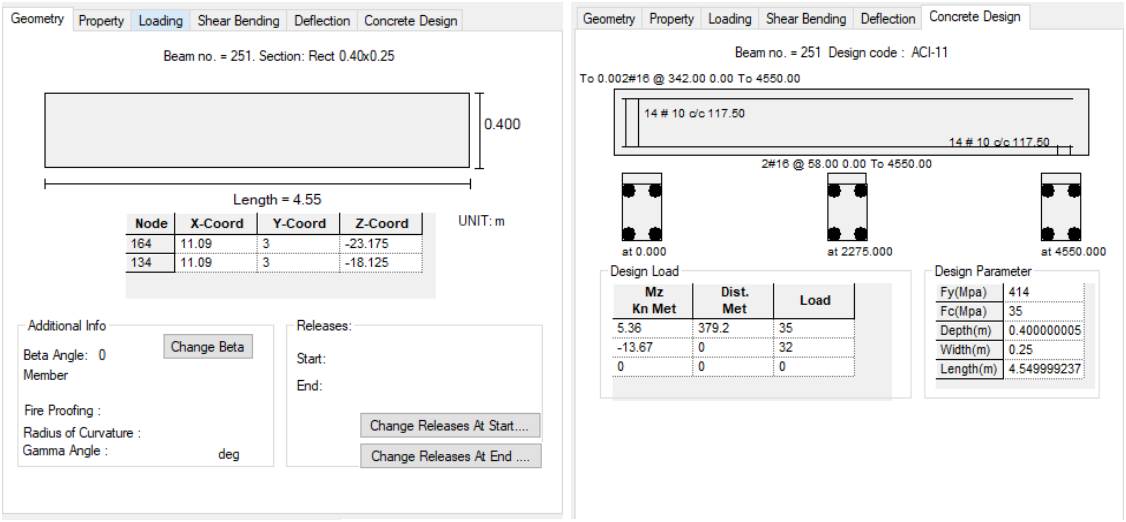
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9.4. Design of Beam-(B1)





Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **303 mm^2**

Provide 2-16 dia-Full Straight Bars Top & Bottom

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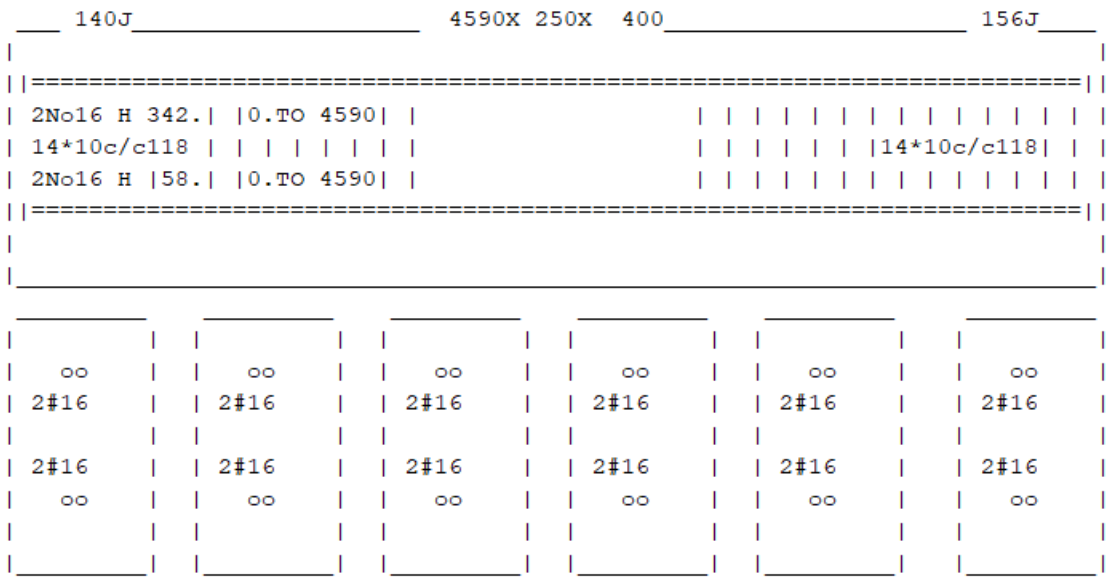
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9.4.1. Staad Output For B1

```
        B E A M  N O.   245 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=     4.86 KNS   Vc=    82.06 KNS   Vs=     0.00 KNS
 Tu=     4.38 KN-MET   Tc=     2.8 KN-MET   Ts=     5.8 KN-MET   LOAD     47
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 118. MM  C/C FOR 1515. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.57 SQ.CM.

 AT END    SUPPORT - Vu=     5.43 KNS   Vc=    80.90 KNS   Vs=     0.00 KNS
 Tu=     8.53 KN-MET   Tc=     2.8 KN-MET   Ts=    11.4 KN-MET   LOAD     25
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 118. MM  C/C FOR 1515. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    3.07 SQ.CM.
```

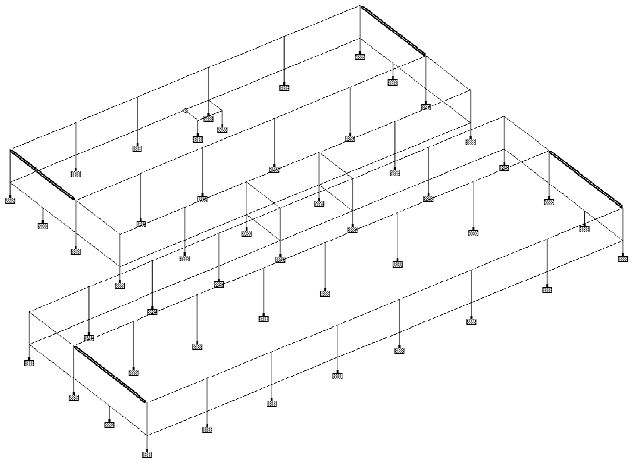


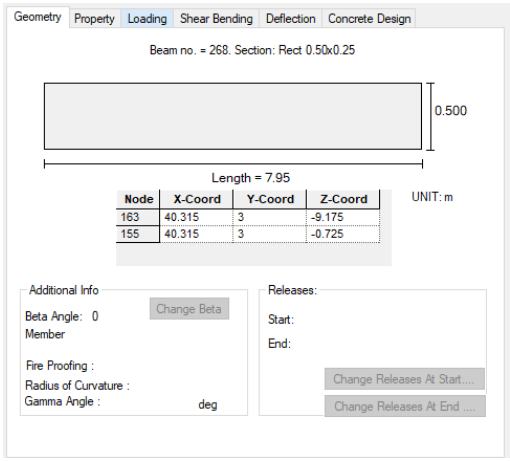
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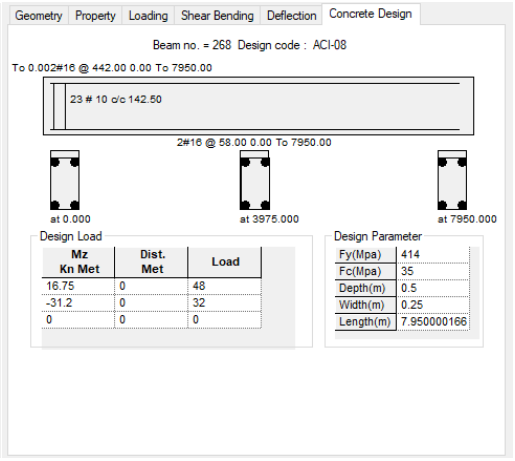
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9.5. Design of Beam-(B2)







Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **482 mm²**

Provide 3-16 dia-Full Straight Bars Top & Bottom

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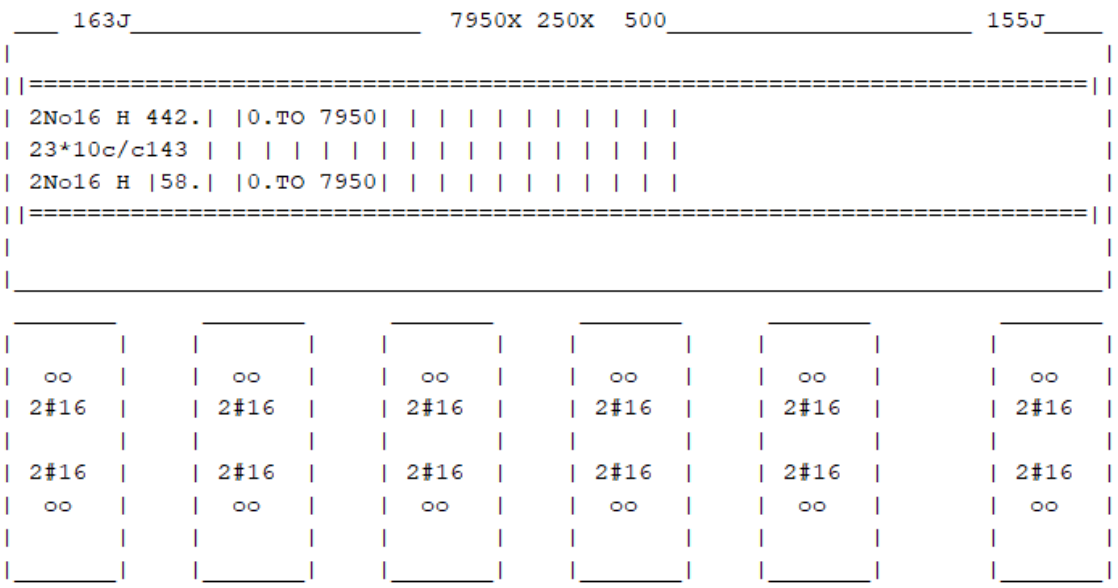
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9.5.1. Staad Output For B2

```
            B E A M  N O.   268 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=   12.93 KNS  Vc=  105.83 KNS  Vs=    0.00 KNS
   Tu=    6.50 KN-MET  Tc=    3.9 KN-MET  Ts=    8.7 KN-MET  LOAD    25
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 3095. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   2.14 SQ.CM.

   AT END   SUPPORT - Vu=    8.98 KNS  Vc=  106.55 KNS  Vs=    0.00 KNS
   Tu=    1.79 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                     STIRRUPS ARE NOT REQUIRED.
```



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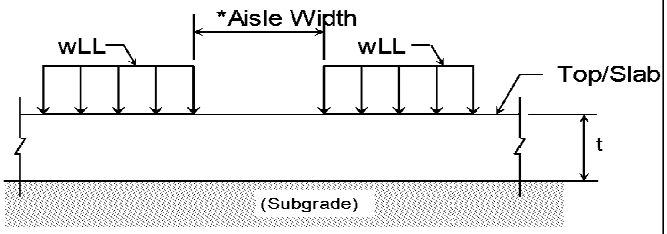
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Low. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) = 682.05 psf = 32.65 kpa

wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)

wLL(allow) >= wLL, O.K.

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

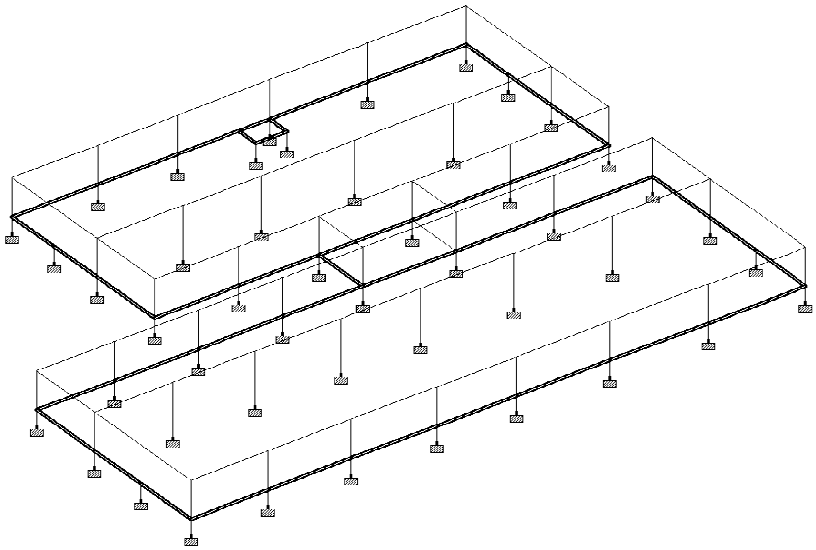
Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

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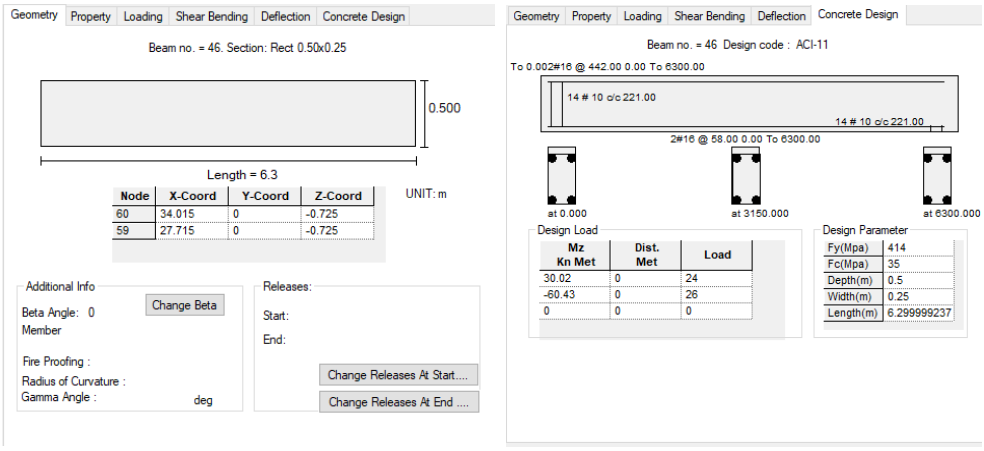
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11. DESIGN OF GRADE BEAM

11.1. Grade Beam(GB1)



MOMENT ENVELOPE FOR BEAMS GB1



Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

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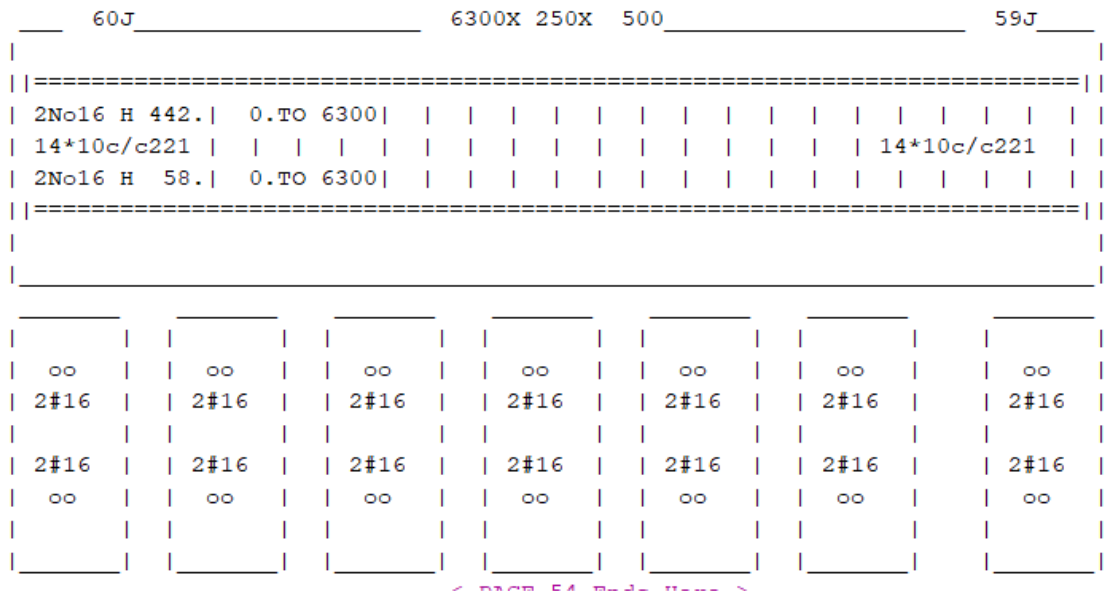
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11.1.1. STAAD OUTPUT FOR GB1

```
                B E A M   N O.    46 D E S I G N   R E S U L T S - SHEAR


 AT START SUPPORT - Vu=    48.95 KNS   Vc=  105.63 KNS   Vs=     0.00 KNS
 Tu=     0.39 KN-MET   Tc=     3.7 KN-MET   Ts=     0.0 KN-MET   LOAD     24
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2710. MM

 AT END    SUPPORT - Vu=    49.28 KNS   Vc=  105.63 KNS   Vs=     0.00 KNS
 Tu=     0.39 KN-MET   Tc=     3.7 KN-MET   Ts=     0.0 KN-MET   LOAD     24
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2710. MM
```

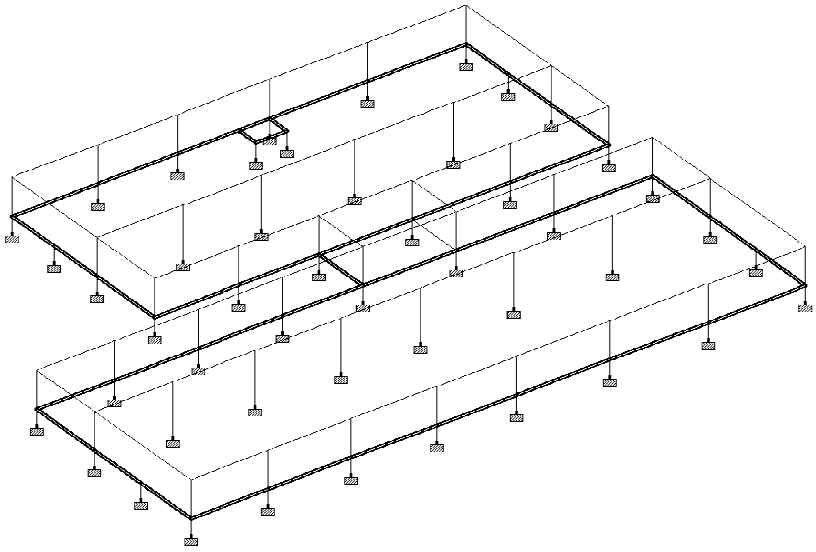


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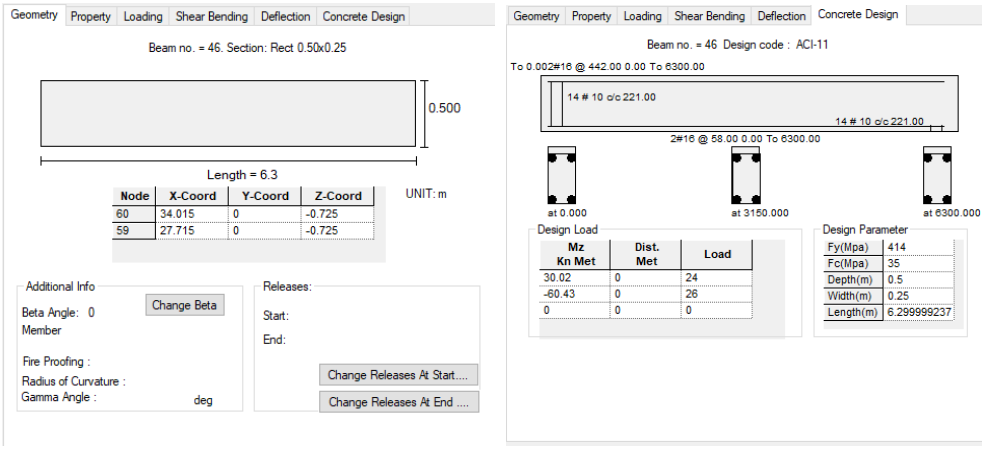
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11.2. Grade Beam(GB2)



MOMENT ENVELOPE FOR BEAMS GB1



Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

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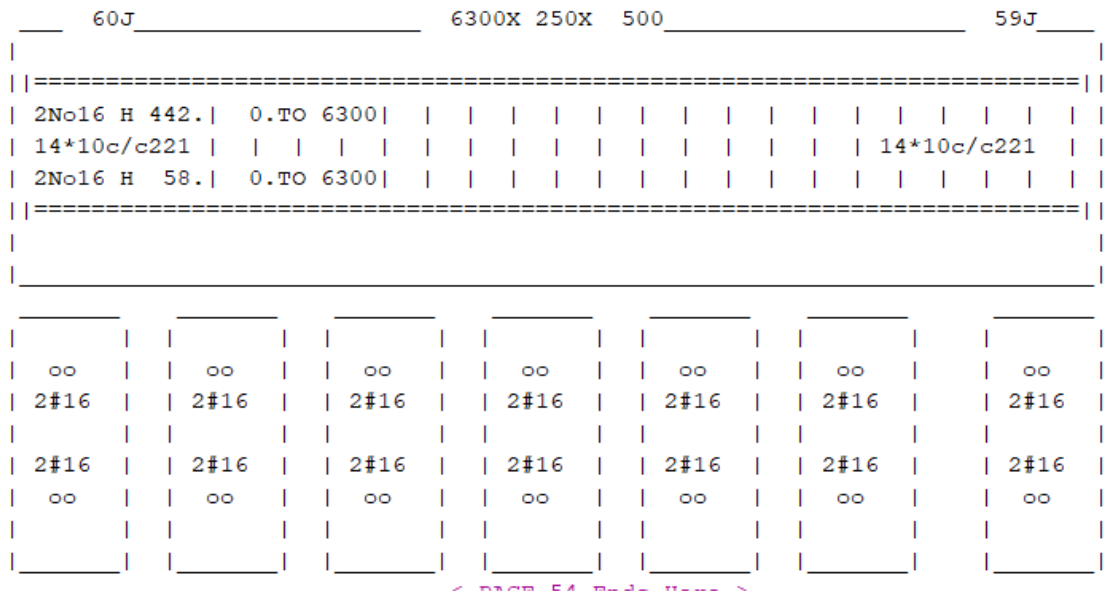
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11.2.1. STAAD OUTPUT FOR GB1

```
            B E A M   N O.    46 D E S I G N   R E S U L T S - SHEAR


 AT START SUPPORT - Vu=    48.95 KNS   Vc=  105.63 KNS   Vs=     0.00 KNS
 Tu=     0.39 KN-MET   Tc=     3.7 KN-MET   Ts=     0.0 KN-MET   LOAD     24
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2710. MM

 AT END    SUPPORT - Vu=    49.28 KNS   Vc=  105.63 KNS   Vs=     0.00 KNS
 Tu=     0.39 KN-MET   Tc=     3.7 KN-MET   Ts=     0.0 KN-MET   LOAD     24
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2710. MM
```

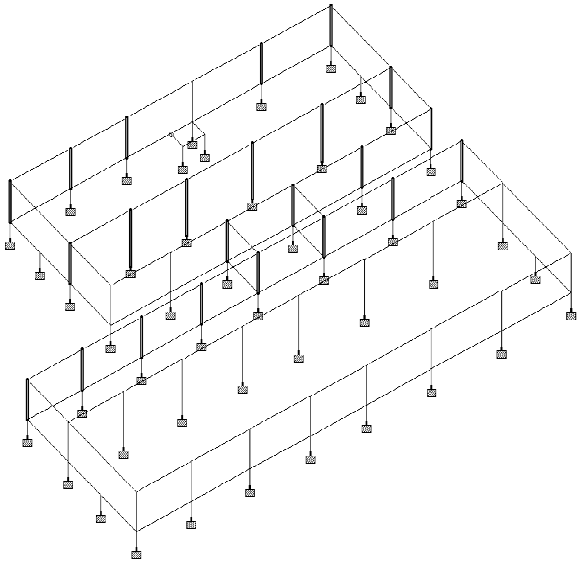


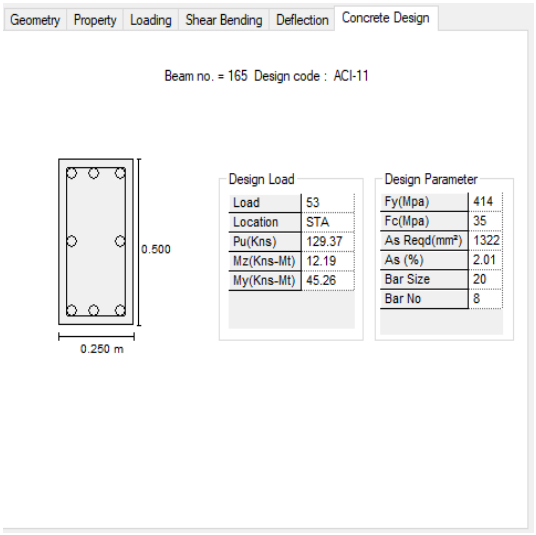
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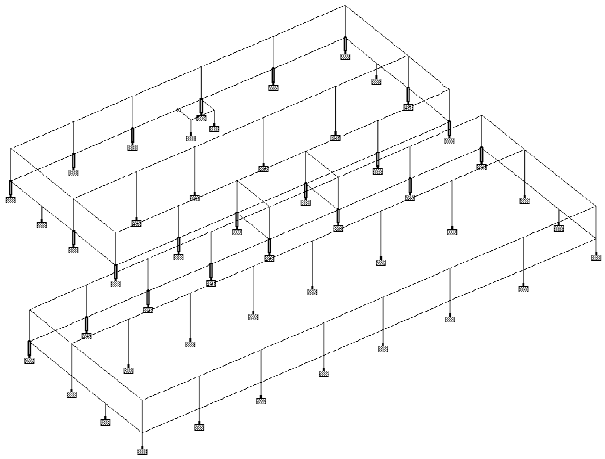
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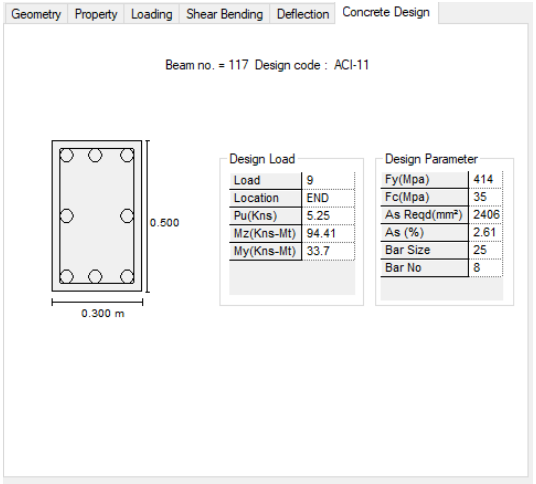
12. CONCRETE COLUMN DESIGN





Column's C1

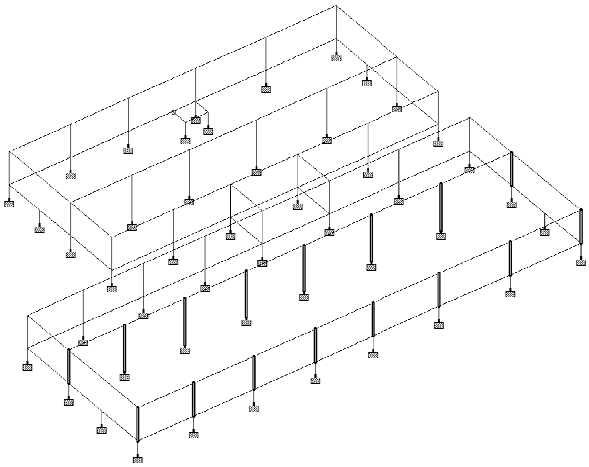


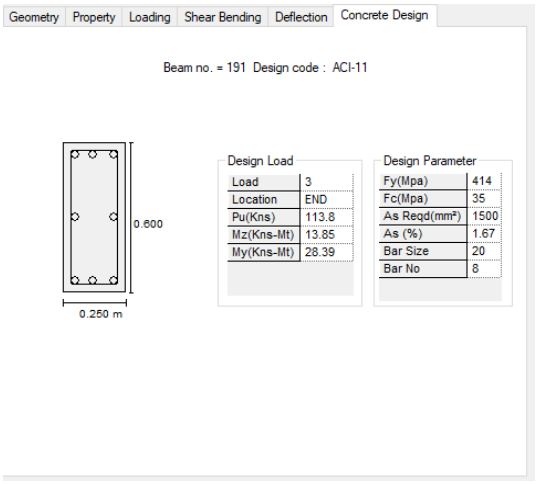


Column's CR1

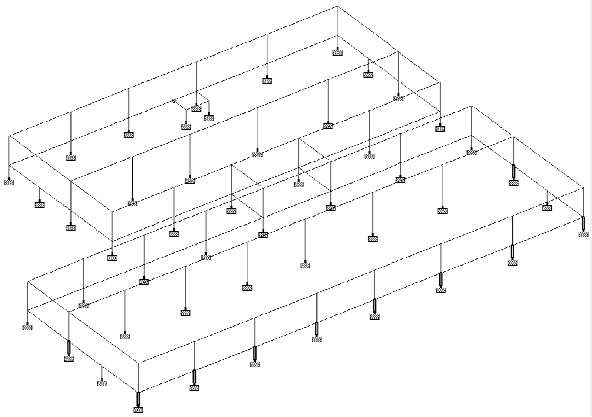
PROPRIETARY & CONFIDENTIAL

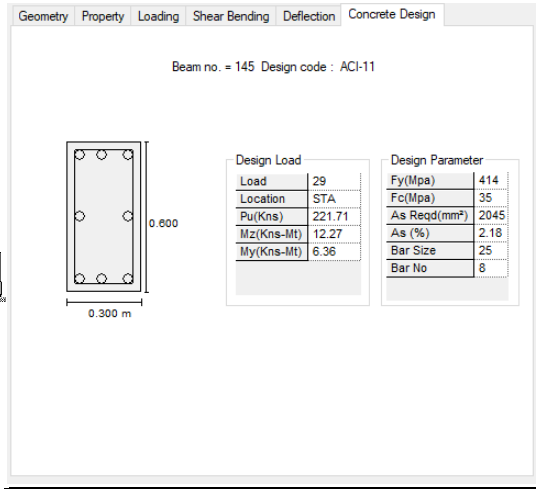
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Column's C2

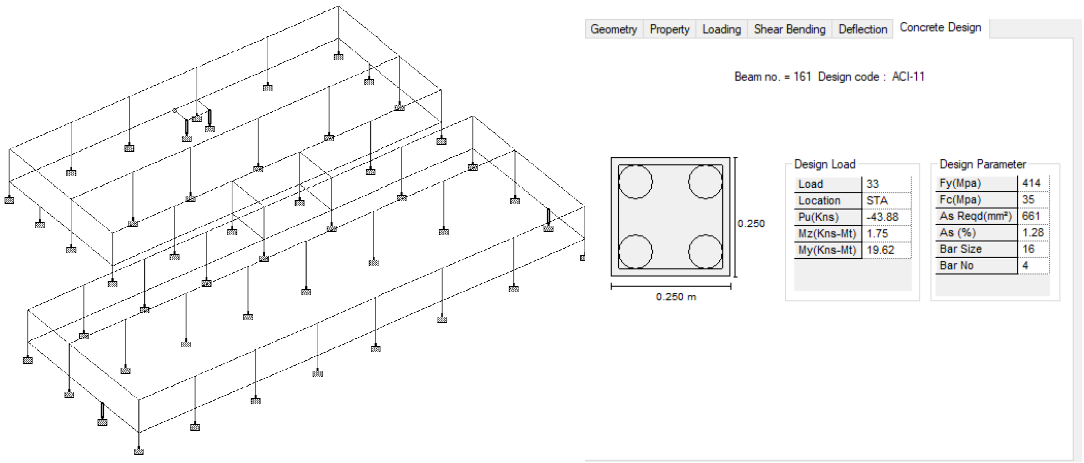




Column's CR2

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Column's CR3

12.1. Staad Output For Column C1 & CR1

```
      ACI 318-11    COLUMN  NO.    165    DESIGN RESULTS
      ================================================


FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 250.0 X 500.0 MMS, TIED
        AREA OF STEEL REQUIRED =   1322.5   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

  8 - 20 MM                2.011        53       STA      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    10 SPACING 250.00 MM
```


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ACI 318-11 COLUMN NO. 117 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 300.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 2406.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	2.618	9	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 300.00 MM

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12.2. Staad Output For Column C2 & CR2

```
      ACI 318-11    COLUMN  NO.    191    DESIGN RESULTS
      ================================================


FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 250.0 X 600.0 MMS, TIED
         ONLY MINIMUM STEEL IS REQUIRED.
         AREA OF STEEL REQUIRED =   1500.0   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

  8 - 20 MM                 1.676         3       END      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    10 SPACING 250.00 MM

      ACI 318-11    COLUMN  NO.    145    DESIGN RESULTS
      ================================================


FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 300.0 X 600.0 MMS, TIED
         AREA OF STEEL REQUIRED =   2044.8   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

  8 - 25 MM                 2.182        29       STA      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    10 SPACING 300.00 MM
```

12.3. Staad Output For Column CR3

```
      ACI 318-11    COLUMN  NO.    161    DESIGN RESULTS
      ================================================


FY - 413.7   FC -   35.0 MPA,   SQRE SIZE - 250.0 X 250.0 MMS, TIED
         AREA OF STEEL REQUIRED =    661.2   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

  4 - 16 MM                 1.287        33       STA      0.900
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    10 SPACING 250.00 MM
```


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13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24		kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm^2
Loading			Modulus of Elasticity of concrete	27806		N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**			
Wall load	0	kN/m^2	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"		468	mm^2
Total Ultimate Load "Uu_{tot}"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"		0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"db" = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**				

Check for Section capacity

Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm					
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	**>**	**3.51**	**kNm**	(Mu)

Shear requirements

Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	**>**	**8.096**	**kN**	(Vu)

Serviceability requirements

d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d').	3.915E+07	mm^4
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm^2
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm^4
Allowable stress fr = 0.62√f'c	3.67	mm^2
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411		
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b **$I_e/360$ 2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = $M_{cr}/Msus$	5.091	
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $I_e/240$ 3.333 mm > 0.079 mm O.K

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14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008-A01 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.) BUILDING PLANS	MD-216-4000-EG-AR-GAD-1075
2.) OVERALL LAYOUT PLAN	MD-216-4000-EG-GAD-1074
3.) LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-4000-EG-CV-DED-1076
4.) LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-4000-EG-CV-DED-1077
5.) GEOTECHNICAL REPORT	MD-216-3000-EG-CV-CAL-1008
6.) STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

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STAAD SPACE DXF IMPORT OF COL & AXES.DXF
START JOB INFORMATION
ENGINEER DATE 03-Feb-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
32 11.09 0 -30.665; 33 16.59 0 -30.665; 34 21.715 0 -30.665;
35 27.715 0 -30.665; 36 34.015 0 -30.665; 37 40.315 0 -30.665;
40 11.09 0 -18.125; 41 16.59 0 -18.125; 42 21.715 0 -18.125;
43 27.715 0 -18.125; 44 34.015 0 -18.125; 45 40.315 0 -18.125;
46 0.724996 0 -14.265; 47 5.725 0 -14.265; 48 11.09 0 -14.265;
49 16.59 0 -14.265; 50 21.715 0 -14.265; 51 27.715 0 -14.265;
52 34.015 0 -14.265; 53 40.315 0 -14.265; 54 0.724996 0 -0.725;
55 5.725 0 -0.725; 56 11.09 0 -0.725; 57 16.59 0 -0.725; 58 21.715 0 -0.725;
59 27.715 0 -0.725; 60 34.015 0 -0.725; 61 40.315 0 -0.725;62 0.724996 0 -9.175; 69
40.315 0 -9.175; 72 11.09 0 -23.175;77 40.315 0 -23.175; 78 0.724996 0 -5.125; 79 40.315
0 -5.125;80 27.715 0 -29.12; 81 25.715 0 -29.12; 82 11.09 -1.2 -30.665;
83 16.59 -1.2 -30.665; 84 21.715 -1.2 -30.665; 85 27.715 -1.2 -30.665;
86 34.015 -1.2 -30.665; 87 40.315 -1.2 -30.665; 88 11.09 -1.2 -18.125;
89 16.59 -1.2 -18.125; 90 21.715 -1.2 -18.125; 91 27.715 -1.2 -18.125;
92 34.015 -1.2 -18.125; 93 40.315 -1.2 -18.125; 94 0.724996 -1.2 -14.265;
95 5.725 -1.2 -14.265; 96 11.09 -1.2 -14.265; 97 16.59 -1.2 -14.265;
98 21.715 -1.2 -14.265; 99 27.715 -1.2 -14.265; 100 34.015 -1.2 -14.265;
101 40.315 -1.2 -14.265; 102 0.724996 -1.2 -0.725; 103 5.725 -1.2 -0.725;
104 11.09 -1.2 -0.725; 105 16.59 -1.2 -0.725; 106 21.715 -1.2 -0.725;
107 27.715 -1.2 -0.725; 108 34.015 -1.2 -0.725; 109 40.315 -1.2 -0.725;
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113 16.59 -1.2 -9.175; 114 21.715 -1.2 -9.175; 115 27.715 -1.2 -9.175;
116 34.015 -1.2 -9.175; 117 40.315 -1.2 -9.175; 118 11.09 -1.2 -23.175;
119 16.59 -1.2 -23.175; 120 21.715 -1.2 -23.175; 121 27.715 -1.2 -23.175;
122 34.015 -1.2 -23.175; 123 40.315 -1.2 -23.175; 124 0.724996 -1.2 -5.125;
125 40.315 -1.2 -5.125; 126 27.715 -1.2 -29.12; 127 25.715 -1.2 -29.12;
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131 27.715 3 -30.665; 132 34.015 3 -30.665; 133 40.315 3 -30.665;
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137 27.715 3 -18.125; 138 34.015 3 -18.125; 139 40.315 3 -18.125;
140 0.724996 3 -14.265; 141 5.725 3 -14.265; 142 11.09 3 -14.265;
143 16.59 3 -14.265; 144 21.715 3 -14.265; 145 27.715 3 -14.265;
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152 21.715 3 -0.725; 153 27.715 3 -0.725; 154 34.015 3 -0.725;
155 40.315 3 -0.725; 156 0.724996 3 -9.175; 157 5.725 3 -9.175;
158 11.09 3 -9.175; 159 16.59 3 -9.175; 160 21.715 3 -9.175;
161 27.715 3 -9.175; 162 34.015 3 -9.175; 163 40.315 3 -9.175;
164 11.09 3 -23.175; 165 16.59 3 -23.175; 166 21.715 3 -23.175;
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170 25.715 0 -30.665; 173 37.815 0 -0.725; 174 11.09 0 -26.925;
175 11.09 -1.2 -26.925; 176 40.315 0 -26.925; 177 40.315 -1.2 -26.925;
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65 62 78; 66 46 62; 80 72 40; 81 32 174; 97 43 51; 99 35 80; 103 69 79;
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169 40 134; 170 41 135; 171 42 136; 172 43 137; 173 44 138; 174 45 139;
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181 52 146; 182 53 147; 183 54 148; 184 55 149; 185 56 150; 186 57 151;
187 58 152; 188 59 153; 189 60 154; 190 61 155; 191 62 156; 192 157 111;
198 69 163; 199 72 164; 204 77 169; 208 129 128; 209 130 129; 210 131 130;
211 132 131; 212 133 132; 213 135 134; 214 136 135; 215 137 136; 216 138 137;
217 139 138; 218 141 140; 219 142 141; 220 143 142; 221 144 143; 222 145 144;
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159 TO 162 291 293 PRIS YD 0.25 ZD 0.25
208 TO 231 265 275 PRIS YD 0.5 ZD 0.25
244 252 268 272 PRIS YD 0.5 ZD 0.25
19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 99 103 104 106 -
107 111 114 115 280 TO 284 289 290 292 PRIS YD 0.5 ZD 0.25
137 TO 145 152 PRIS YD 0.6 ZD 0.3
MEMBER PROPERTY AMERICAN
154 TO 157 163 TO 182 199 204 PRIS YD 0.5 ZD 0.25
117 TO 136 153 158 PRIS YD 0.5 ZD 0.3
245 251 269 271 PRIS YD 0.4 ZD 0.25
MEMBER PROPERTY AMERICAN
232 TO 243 PRIS YD 0.55 ZD 0.25
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ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
CONSTANTS
BETA 90 MEMB 147 150 153 156 199 291 293
MATERIAL CONCRETE ALL
SUPPORTS
82 TO 127 175 177 FIXED
MEMBER RELEASE
282 END MX MY MZ
MEMBER OFFSET
208 TO 211 213 TO 216 218 TO 223 225 TO 237 239 TO 242 START -0.25 0 0
208 TO 211 213 214 216 TO 231 233 TO 235 237 238 240 243 END 0.25 0 0
244 245 251 252 265 268 269 271 272 275 START 0 0 0.25
244 245 251 252 265 268 269 271 272 275 END 0 0 -0.25
212 224 238 243 START -0.125 0 0
215 232 236 239 241 END 0.125 0 0
MEMBER CRACKED
117 TO 145 147 TO 150 152 TO 192 198 199 -
204 REDUCTION RIY 0.700000 RIZ 0.700000
19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 99 103 104 106 -

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107 111 114 115 208 TO 245 251 252 265 268 269 271 272 275 280 TO 284 -
289 REDUCTION RIY 0.350000 RIZ 0.350000
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2
SELFWEIGHT 1
MEMBER WEIGHT
117 TO 145 208 TO 212 UNI 24
117 TO 145 225 TO 231 UNI 28.43
213 TO 224 UNI 14.9
232 TO 238 UNI 43.32
239 TO 243 UNI 38.9
265 275 UNI 19.2
19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 99 103 104 106 -
107 111 114 115 117 TO 145 152 153 158 TO 162 280 TO 282 289 TO 293 UNI 10
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
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1 4.000 M 40.000 M 30.000 M 2.000 0.010 0 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.871 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.61878 1.61878 HEIG  0  4.572
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 40.000 M 30.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.871 -0.433 0.550
!> END GENERATED DATA BLOCK
INT -1.20414 -1.20414 HEIG  0  4
TYPE 3 WIND X SIDE WALL
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 40.000 M 30.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.48663 -1.48663 HEIG  0  4
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 30.000 M 40.000 M 2.000 0.010 0 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 HEIG  0  4.572
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 30.000 M 40.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.863 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.27428 -1.27428 HEIG  0  4
TYPE 6 WIND Z SIDE WALL
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 30.000 M 40.000 M 2.000 0.010 2 -
```

```
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.48663 -1.48663 HEIG  0  4
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
117 TO 145 208 TO 212 UNI GY -24
MEMBER LOAD
117 TO 145 225 TO 231 UNI GY -28.43
117 244 245 251 252 275 UMOM GZ 1
117 265 268 269 271 272 UMOM GZ -1
117 208 TO 212 218 TO 224 UMOM GX -1
117 213 214 216 217 222 225 TO 231 UMOM GX 1
213 TO 224 UNI GY -14.9
232 TO 238 UNI GY -43.32
239 TO 243 UNI GY -38.9
265 275 UNI GY -19.2
19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 99 103 104 106 -
107 111 114 115 117 TO 145 152 153 158 TO 162 280 TO 282 289 TO 292 -
293 UNI GY -10
LOAD 4 LOADTYPE Live  TITLE LL
MEMBER LOAD
208 TO 212 UNI GY -1.875
225 TO 231 UNI GY -2.1
213 TO 224 UNI GY -1.275
232 TO 238 UNI GY -3.385
232 TO 243 UNI GY -3.15
244 245 251 252 275 UMOM GZ 0.1
265 268 269 271 272 UMOM GZ -0.1
208 TO 212 218 TO 221 223 224 UMOM GX -0.1
213 214 216 217 225 TO 231 UMOM GX 0.1
265 275 UNI GY -1.5
LOAD 5 LOADTYPE Wind  TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 4
WIND LOAD -X 1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
WIND LOAD -Z -1 TYPE 3 YR 0 4
LOAD 6 LOADTYPE Wind  TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 4
WIND LOAD -Z 1 TYPE 5 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
LOAD 7 LOADTYPE Wind  TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 4
WIND LOAD -X -1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
WIND LOAD -Z -1 TYPE 3 YR 0 4
LOAD 8 LOADTYPE Wind  TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 4
WIND LOAD -Z -1 TYPE 5 YR 0 4
```

WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
LOAD 9 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 99 103 104 106 -
107 111 114 115 147 TO 151 154 TO 157 163 TO 192 198 199 204 208 TO 245 251 -
252 265 268 269 271 272 275 280 TO 282 TEMP 25
LOAD COMB 10 SLS: (DL)
3 1.0
LOAD COMB 11 SLS: (DL+T)
3 1.0
LOAD COMB 12 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 13 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 14 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 15 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 16 SLS: (D+0.75WX1+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 17 SLS: (D+0.75WZ1+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 18 SLS: (0.6D+WX1)
3 0.6 5 1.0
LOAD COMB 19 SLS: (0.6D+WZ1)
3 0.6 6 1.0
LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)
3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 25 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 26 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 27 ULS: (1.2D+0.80WX1+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 29 ULS: (1.2D+1.6WX1+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 30 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0

LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 39 SLS: (D+0.75WX2+0.75LR)
3 1.0 7 0.75 4 0.75
LOAD COMB 40 SLS: (D+0.75WZ2+0.75LR)
3 1.0 8 0.75 4 0.75
LOAD COMB 41 SLS: (0.6D+WX2)
3 0.6 7 1.0
LOAD COMB 42 SLS: (0.6D+WZ2)
3 0.6 8 1.0
LOAD COMB 45 SLS: (D+WX2)
3 1.0 7 1.0
LOAD COMB 46 SLS: (D+WZ2)
3 1.0 8 1.0
LOAD COMB 47 ULS: (1.2D+0.80WX2+1.6LR)
3 1.2 7 0.8 4 1.6
LOAD COMB 48 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 8 1.6 4 0.5
LOAD COMB 49 ULS: (1.2D+1.6WX2+0.5LR)
3 1.2 4 0.5 7 1.6
LOAD COMB 50 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 4 0.5 8 1.6
LOAD COMB 53 ULS: (0.90D+1.6WX2)
3 0.9 7 1.6
LOAD COMB 54 ULS: (0.90D+1.6WZ2)
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 MEMB 19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 -
99 103 104 106 107 111 114 115 117 TO 145 147 TO 150 152 TO 192 198 199 204 -
208 TO 245 251 252 265 268 269 271 272 275 280 TO 282 289 TO 293
CLB 0.04 MEMB 19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 -
99 103 104 106 107 111 114 115 117 TO 145 147 TO 150 152 TO 192 198 199 204 -
208 TO 245 251 252 265 268 269 271 272 275 280 TO 282 289 TO 293
CLS 0.04 MEMB 19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 -
99 103 104 106 107 111 114 115 117 TO 145 147 TO 150 152 TO 192 198 199 204 -
208 TO 245 251 252 265 268 269 271 272 275 280 TO 282 289 TO 293
CLT 0.04 MEMB 19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 -
99 103 104 106 107 111 114 115 117 TO 145 147 TO 150 152 TO 192 198 199 204 -
208 TO 245 251 252 265 268 269 271 272 275 280 TO 282 289 TO 293
MINSEC 10 MEMB 19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 -
99 103 104 106 107 111 114 115 117 TO 145 147 TO 150 152 TO 192 198 199 204 -
208 TO 245 251 252 265 268 269 271 272 275 280 TO 282 289 TO 293
SFACE 0.44 MEMB 208 TO 245 251 252 265 268 269 271 272 275
EFACE 0.44 MEMB 208 TO 245 251 252 265 268 269 271 272 275
MAXMAIN 25 MEMB 117 TO 145 152 153 158 TO 162 208 TO 243 265 275 291 293
MINMAIN 25 MEMB 117 TO 145 152 153 158 TO 162 198 243 291 293
MINMAIN 20 MEMB 163 TO 191 198 199 204 243
MAXMAIN 16 MEMB 19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 -
99 103 104 106 107 111 114 115 244 245 251 252 268 269 271 272 280 TO 284 -
289 290 292
MINMAIN 16 MEMB 19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 65 66 80 81 97 -
99 103 104 106 107 111 114 115 159 TO 162 208 TO 245 251 252 265 268 269 -
271 272 275 280 TO 284 289 290 292

```
MAXMAIN 20 MEMB 163 TO 191 198 199 204
DESIGN BEAM 65 66 80 81 97 99 103 104 106 107 111 114 115 244 245 251 252 -
265 268 269 271 272 275 282 290 292
DESIGN BEAM 19 TO 23 27 28 30 31 33 TO 36 38 39 41 TO 47 208 TO 243 280 281-
283 284 289
DESIGN COLUMN 117 TO 145 147 TO 192 198 199 204 291 293
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

PROPRIETARY & CONFIDENTIAL

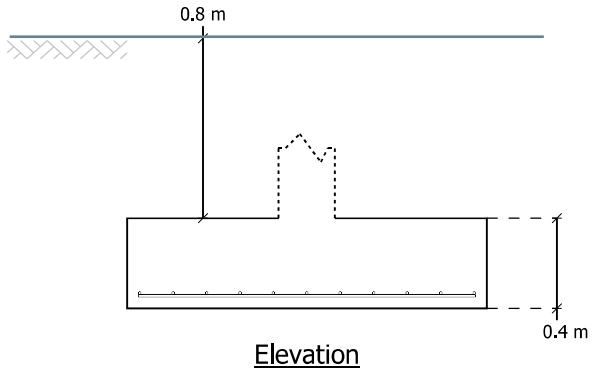
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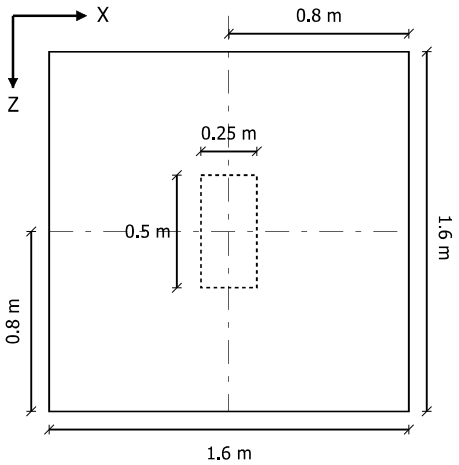
FOUNDATION-F1-GRID 5 & F

Footing No.	Group ID	Foundation Geometry		
		Length	Width	Thickness
-	-			

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
-						



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension

Footing Thickness (Ft) : 400.000mm

Footing Length - X (Fl) : 1600.000mm

Footing Width - Z (Fw) : 1600.000mm

Eccentricity along X (Oxd) : 0.000mm

Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular

Column Length - X (D_{col}) : 0.250m

Column Width - Z (B_{col}) : 0.500m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	24.000kN/m3
Strength of Concrete :	35.000N/mm2
Yield Strength of Steel :	420.000N/mm2
Minimum Bar Size :	#12
Maximum Bar Size :	#12
Top Footing Minimum Bar Size :	#12
Top Footing Maximum Bar Size :	#12
Pedestal Minimum Bar Size :	#16
Pedestal Maximum Bar Size :	#16
Minimum Bar Spacing :	100.000mm
Maximum Bar Spacing :	150.000mm
Pedestal Clear Cover (P, CL) :	50.000mm
Bottom Footing Clear Cover (F, CL) :	50.000mm

Soil Properties

Soil Type :	Cohesionless Soil
Unit Weight :	18.000kN/m3
Soil Bearing Capacity :	570.000kN/m2
Soil Bearing Capacity Type:	Net Bearing Capacity
Soil Surcharge :	0.000kN/m2
Depth of Soil above Footing :	800.000mm
Type of Depth :	Fixed Top
Undrained Shear Strength :	0.000kN/m2
Bearing Capacity Input Method:	Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction :	0.580
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Global Settings

Top Reinforcement Option :	Calculate only when foundation is subjected to uplift forces
Concrete Design Option :	Gross Pressure
Top Reinforcement Factor :	1.000

Design Calculations

Footing Size

Initial Length (L_o) =	1.600m
Initial Width (W_o) =	1.600m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	274.195	0.268	0.019	0.144	-0.310
12	311.184	0.308	0.018	0.143	-0.359
13	301.937	0.298	0.018	0.143	-0.346
14	274.190	0.266	17.079	74.511	-0.305
15	273.753	5.054	0.031	0.241	-10.309
16	301.753	1.780	0.002	0.108	-3.439
17	301.838	0.278	0.001	0.090	-0.300
18	164.272	2.137	-0.011	0.039	-4.309
19	164.385	0.134	-0.012	0.016	-0.124
20	164.075	4.947	0.024	0.183	-10.184
21	164.512	0.158	17.072	74.454	-0.180
22	273.950	2.244	-0.004	0.097	-4.433
23	274.063	0.242	-0.005	0.073	-0.249

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
39	302.149	-1.037	-0.004	0.066	2.436
40	301.815	0.463	0.041	0.214	-0.687
41	164.800	-1.618	-0.019	-0.017	3.524
42	164.355	0.381	0.041	0.181	-0.641
45	274.478	-1.511	-0.011	0.041	3.400
46	274.033	0.489	0.048	0.239	-0.765

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	383.873	0.375	0.026	0.201	-0.435
25	402.367	0.395	0.026	0.201	-0.459
26	402.367	0.395	0.026	0.201	-0.459
27	388.020	1.966	0.004	0.134	-3.748
28	347.317	0.299	-0.015	0.059	-0.298
29	347.136	3.503	-0.014	0.097	-6.993
30	347.317	0.299	-0.015	0.059	-0.298
31	328.402	7.159	0.040	0.311	-14.656
32	329.027	0.318	24.394	106.412	-0.364
33	246.383	3.403	-0.019	0.054	-6.876
34	246.564	0.199	-0.020	0.016	-0.180
35	246.768	0.238	24.389	106.369	-0.271
36	246.144	7.079	0.035	0.268	-14.563
47	388.443	-1.038	-0.002	0.089	2.519
48	347.269	0.695	0.070	0.324	-1.124
49	347.981	-2.505	-0.025	0.007	5.540
50	347.269	0.695	0.070	0.324	-1.124
53	247.228	-2.605	-0.031	-0.036	5.657
54	246.516	0.594	0.064	0.281	-1.006

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.560 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.083 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.600 m Governing Load Case : # 11

Width (W_2) = 1.600 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 32

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.560 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 24.575 kN

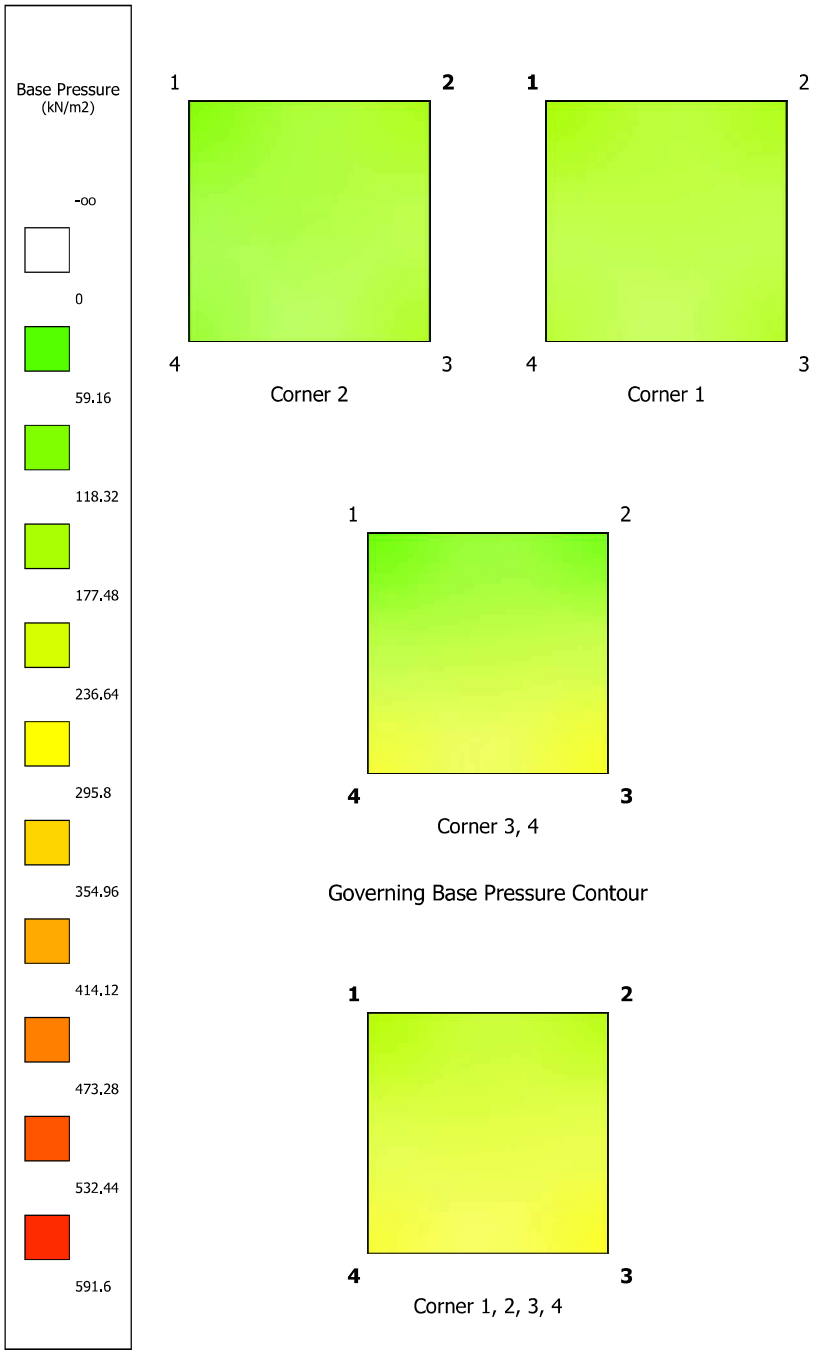
Gross Soil Bearing Capacity = 591.60 kN/m2

Soil Weight On Top Of Footing = 35.063 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

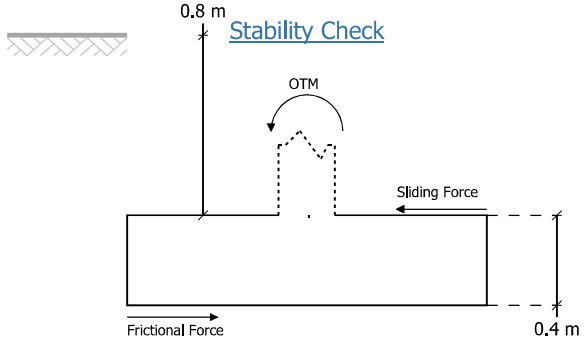


Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
39	**145.4044**	137.0526	137.2416	145.5934	0.000
15	111.7978	**147.9225**	148.6637	112.5390	0.000
14	10.6431	11.8468	**250.1599**	248.9562	0.000
14	10.6431	11.8468	250.1599	**248.9562**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
39	**145.4044**	137.0526	137.2416	145.5934
15	111.7978	**147.9225**	148.6637	112.5390
14	10.6431	11.8468	**250.1599**	248.9562
14	10.6431	11.8468	250.1599	**248.9562**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	722.008	10414.318	1018.628	1.500	1764.928	641.873	1.500
12	698.750	11666.760	986.415	1.500	1969.869	618.184	1.500
13	703.984	11351.808	993.675	1.500	1918.458	623.497	1.500
14	729.230	11.337	16.031	1.500	3.283	652.607	1.500
15	38.258	6193.084	54.104	1.500	1054.223	21.718	1.500
16	117.764	129227.293	166.543	1.500	2661.302	69.914	1.500
17	754.310	188539.437	1066.746	1.500	3185.949	705.720	1.500
18	60.774	11478.489	85.946	1.500	5176.786	34.896	1.500
19	967.470	10833.185	1362.786	1.500	16628.640	1011.973	1.500
20	26.229	5455.035	37.093	1.500	929.876	14.803	1.500
21	821.560	7.615	10.769	1.500	2.206	740.024	1.500
22	86.215	49904.773	121.926	1.500	2805.321	50.261	1.500
23	801.199	42472.334	1132.865	1.500	3743.897	776.328	1.500
39	202.443	54347.685	286.295	1.500	4487.154	101.907	1.500
40	452.491	5132.140	637.446	1.500	1253.248	332.609	1.500
41	80.438	6989.582	113.749	1.500	7406.302	43.304	1.500
42	340.580	3169.606	478.896	1.500	907.793	227.255	1.500
45	128.247	17320.984	181.363	1.500	7366.570	67.024	1.500
46	395.988	3996.469	557.283	1.500	1034.949	279.076	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20

Governing Disturbing Force : 4.947kN

Governing Restoring Force : 129.753kN

Minimum Sliding Ratio for the Critical Load Case : 26.229

Critical Load Case for Overturning about X-Direction : 21

Governing Overturning Moment : 81.282kNm

Governing Resisting Moment : 179.317kNm

Minimum Overturning Ratio for the Critical Load Case : 2.206

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 21

Governing Disturbing Force : 17.072kN

Governing Restoring Force : 130.007kN

Minimum Sliding Ratio for the Critical Load Case : 7.615

Critical Load Case for Overturning about Z-Direction : 20

Governing Overturning Moment : -12.163kNm

Governing Resisting Moment : 180.047kNm

Minimum Overturning Ratio for the Critical Load Case : 14.803

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 21

Governing Disturbing Force : 17.072kN

Governing Restoring Force : 183.858kN

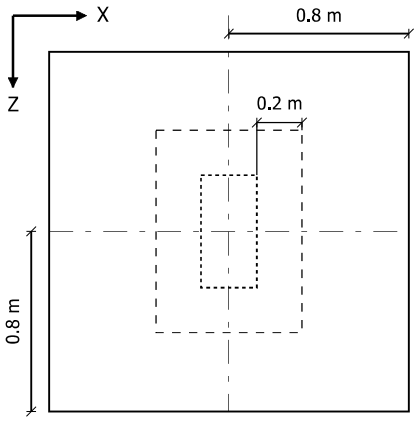
Minimum Sliding Ratio for the Critical Load Case : 10.769

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 371.529kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta_c}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

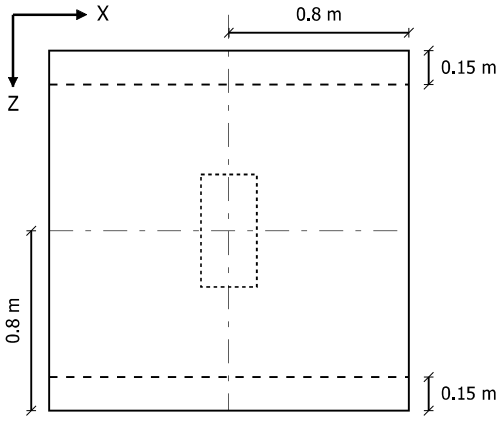
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 540.757kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.206m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

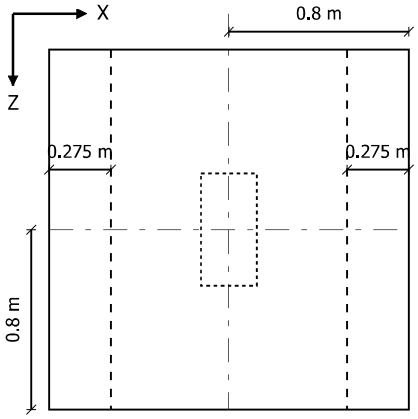
From above calculations, 0.75 X V_c = 405.568 kN

Critical load case for V_{ux} is # 32 $V_{ux} = V_{ux}|_{x=D_x}$ = 97.749 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 540.757 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.331 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

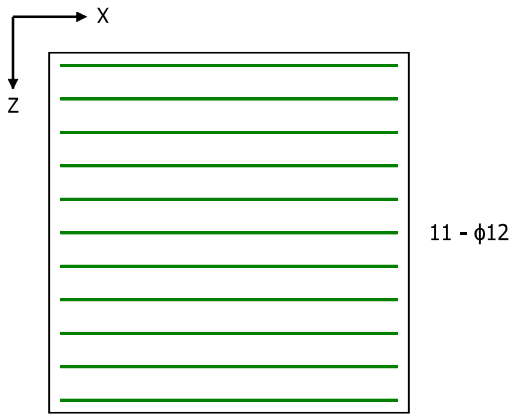
From above calculations, 0.75 X V_c = 405.568 kN

Critical load case for V_{uz} is # 25 $V_{uz} = V_{uz}|_{z=D_z}$ = 95.957 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00172
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.925 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	66.017 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	73.353 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00093
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00080
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1098.176 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.625 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 11

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1244.062 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00226

From ACI Cl.7.6.1, minimum req'd clear distance between bars

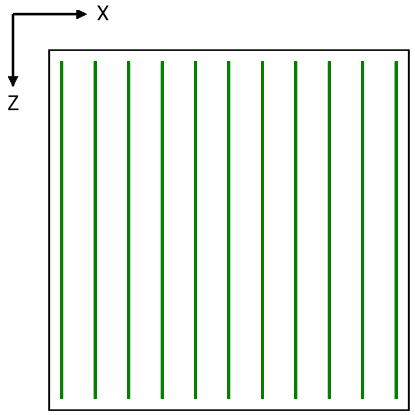
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.133

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



11 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 35

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00168

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.050 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 69.991 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 77.768 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00106

(Based on gross depth) ρ x d_{eff} / Depth = 0.00088

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1076.374 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.500 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 11

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1244.062 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

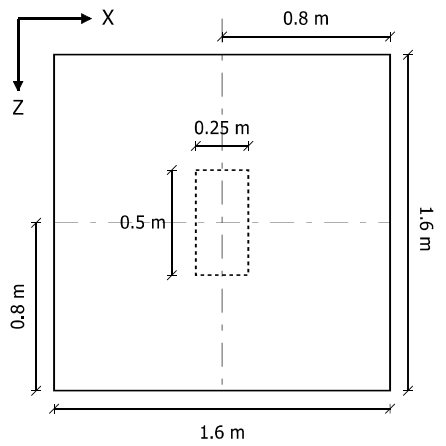
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00234

From ACI Cl.7.6.1, minimum req'd clear distance between bars

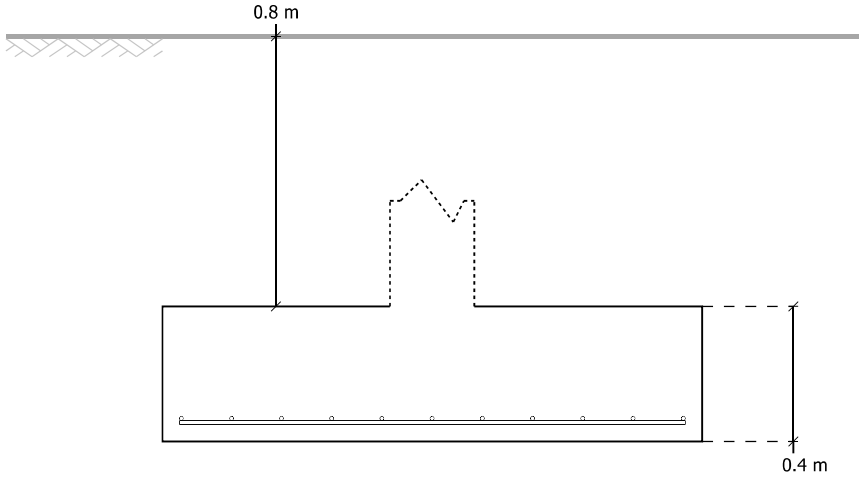
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.156

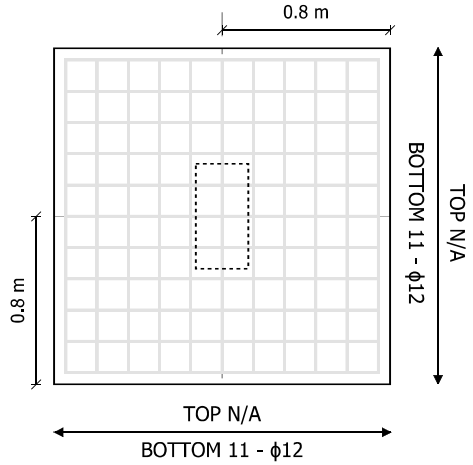
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

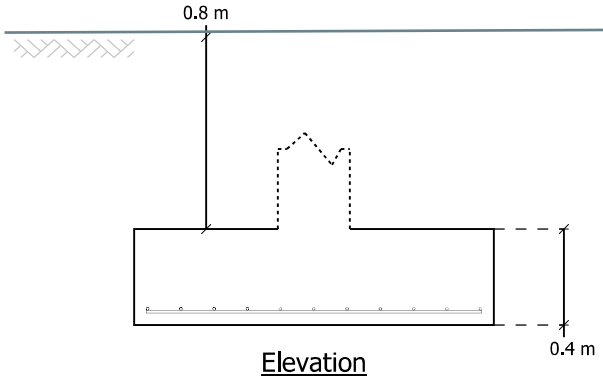
	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	1.6 m	1.6 m	0.4 m	N/A	N/A	11 - ϕ12	11 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

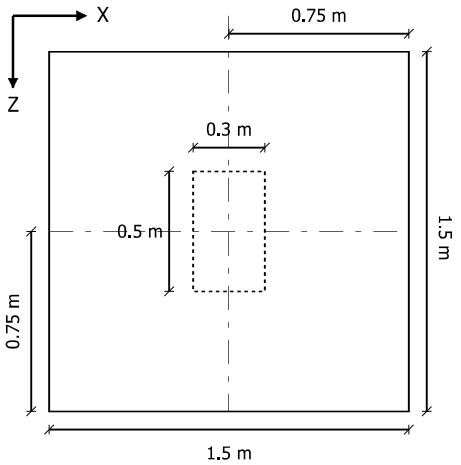
FOUNDATION-F2-GRID 2 & C

Footing No.	Group ID	Foundation Geometry		
		Length	Width	Thickness
-	-			

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
-						



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1500.000mm
Footing Width - Z (Fw) : 1500.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m
Column Width - Z (B_{col}) : 0.500m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	24.000kN/m3
Strength of Concrete :	35.000N/mm2
Yield Strength of Steel :	420.000N/mm2
Minimum Bar Size :	#12
Maximum Bar Size :	#12
Top Footing Minimum Bar Size :	#12
Top Footing Maximum Bar Size :	#12
Pedestal Minimum Bar Size :	#16
Pedestal Maximum Bar Size :	#16
Minimum Bar Spacing :	100.000mm
Maximum Bar Spacing :	150.000mm
Pedestal Clear Cover (P, CL) :	50.000mm
Bottom Footing Clear Cover (F, CL) :	50.000mm

Soil Properties

Soil Type :	Cohesionless Soil
Unit Weight :	18.000kN/m3
Soil Bearing Capacity :	570.000kN/m2
Soil Bearing Capacity Type:	Net Bearing Capacity
Soil Surcharge :	0.000kN/m2
Depth of Soil above Footing :	800.000mm
Type of Depth :	Fixed Top
Undrained Shear Strength :	0.000kN/m2
Bearing Capacity Input Method:	Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction :	0.580
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Global Settings

Top Reinforcement Option :	Calculate only when foundation is subjected to uplift forces
Concrete Design Option :	Gross Pressure
Top Reinforcement Factor :	1.000

Design Calculations

Footing Size

Initial Length (L_o) =	1.500m
Initial Width (W_o) =	1.500m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	222.511	-10.293	21.971	6.514	15.423
12	229.305	-11.049	21.974	6.499	16.585
13	227.607	-10.860	21.973	6.503	16.294
14	219.531	-10.321	32.636	20.945	15.472
15	217.848	-3.140	21.864	6.347	1.417
16	226.232	1.612	16.160	-0.788	2.466
17	227.131	-20.146	28.183	14.279	24.284
18	131.674	10.454	5.431	-5.813	-9.184
19	132.873	-18.557	21.462	14.276	19.907
20	128.843	0.977	13.076	3.741	-4.752
21	130.526	-6.204	23.847	18.339	9.303
22	220.679	6.337	14.220	-3.207	-3.015
23	221.877	-22.675	30.250	16.882	26.076

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
39	229.491	-20.400	16.237	-0.670	27.535
40	227.560	-19.757	17.031	0.313	23.876
41	136.020	-18.896	5.534	-5.655	24.241
42	133.445	-18.039	6.593	-4.344	19.363
45	225.024	-23.013	14.322	-3.050	30.411
46	222.449	-22.156	15.381	-1.739	25.532

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	311.515	-14.410	30.759	9.120	21.592
25	314.913	-14.788	30.761	9.112	22.173
26	314.913	-14.788	30.761	9.112	22.173
27	276.418	-0.257	20.169	0.015	5.616
28	269.397	-32.540	39.614	24.398	36.133
29	267.478	13.878	13.965	-7.745	-10.413
30	269.397	-32.540	39.614	24.398	36.133
31	260.351	-2.133	26.212	7.578	-1.501
32	262.756	-12.391	41.600	28.432	18.578
33	197.328	17.344	7.372	-9.692	-15.620
34	199.246	-29.074	33.021	22.451	30.925
35	196.003	-9.303	35.009	26.478	13.951
36	193.598	0.955	19.621	5.623	-6.128
47	279.894	-23.737	20.251	0.142	32.357
48	270.311	-31.710	15.823	-5.395	35.263
49	274.431	-33.081	14.129	-7.493	43.069
50	270.311	-31.710	15.823	-5.395	35.263
53	204.281	-29.615	7.536	-9.439	37.861
54	200.161	-28.244	9.230	-7.342	30.056

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.250 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.896 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.500 m Governing Load Case : # 11

Width (W_2) = 1.500 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.250 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 21.599 kN

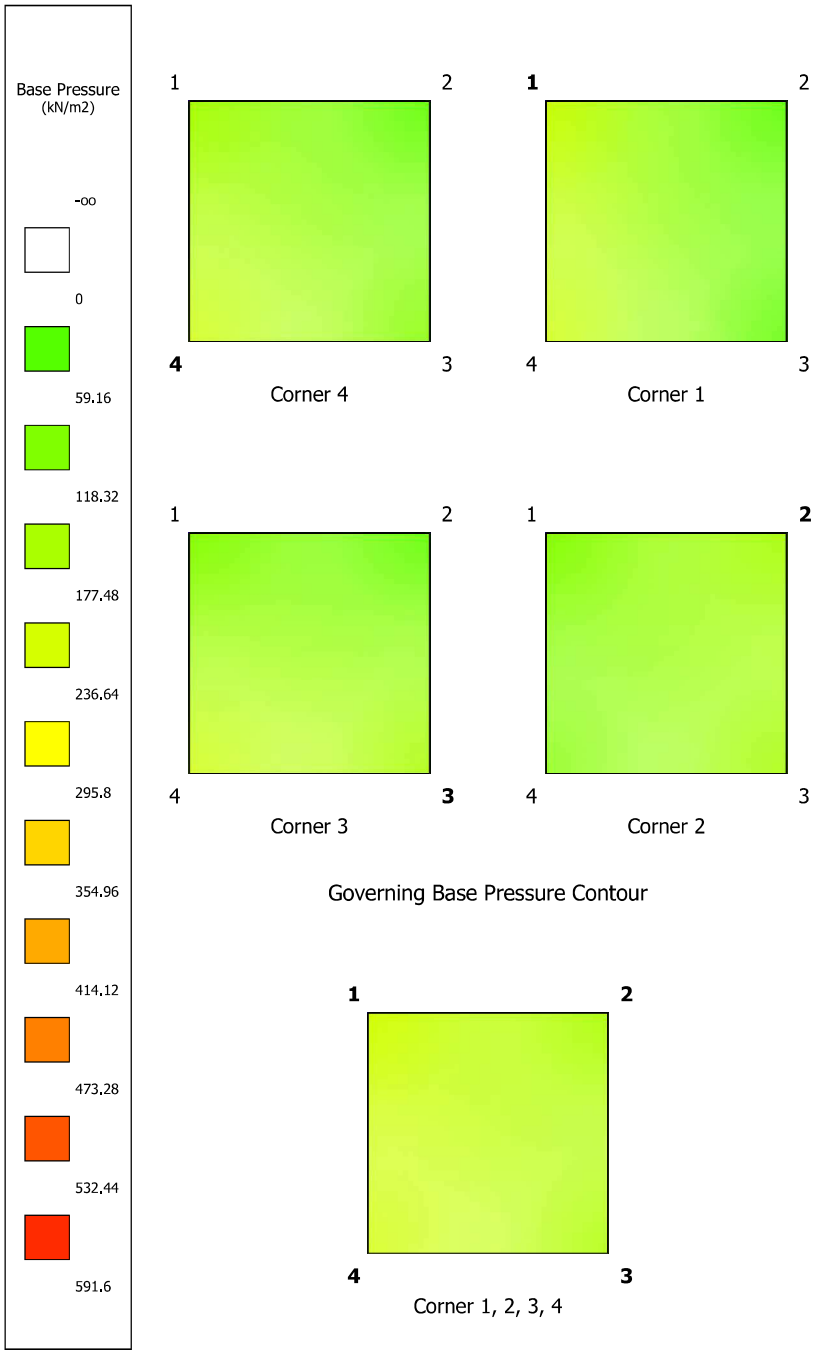
Gross Soil Bearing Capacity = 591.60 kN/m2

Soil Weight On Top Of Footing = 30.239 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

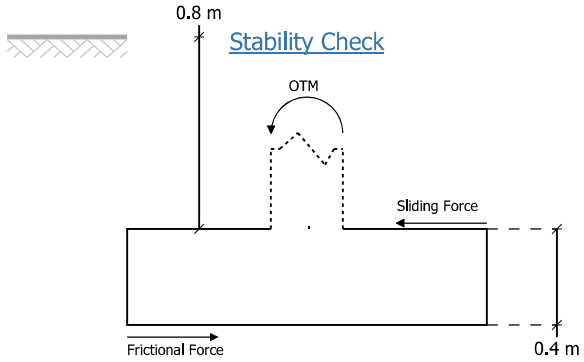


Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**188.7157**	47.8579	57.3840	198.2418	0.000
22	106.8425	**126.5756**	135.3947	115.6616	0.000
14	95.0108	25.3200	**146.2061**	215.8970	0.000
23	132.6091	7.6447	110.6937	**235.6581**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**188.7157**	47.8579	57.3840	198.2418
22	106.8425	**126.5756**	135.3947	115.6616
14	95.0108	25.3200	**146.2061**	215.8970
23	132.6091	7.6447	110.6937	**235.6581**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	15.459	7.242	9.275	1.500	13.464	10.588	1.500
12	14.758	7.421	9.376	1.500	13.809	10.093	1.500
13	14.924	7.376	9.352	1.500	13.723	10.210	1.500
14	15.250	4.823	6.503	1.500	5.994	10.442	1.500
15	49.818	7.154	10.015	1.500	13.420	76.092	1.500
16	100.023	9.980	14.045	1.500	36.794	115.159	1.500
17	8.031	5.741	6.605	1.500	8.199	6.504	1.500
18	10.182	19.597	12.777	1.500	37.878	10.382	1.500
19	5.773	4.992	5.340	1.500	6.072	5.110	1.500
20	107.212	8.015	11.303	1.500	15.135	26.568	1.500
21	17.050	4.435	6.071	1.500	4.916	11.703	1.500
22	24.944	11.116	14.359	1.500	82.511	37.031	1.500
23	7.001	5.248	5.939	1.500	7.092	5.873	1.500
39	7.999	10.049	8.851	1.500	36.271	5.943	1.500
40	8.202	9.515	8.786	1.500	29.446	6.629	1.500
41	5.766	19.689	7.826	1.500	41.013	4.466	1.500
42	5.957	16.300	7.913	1.500	81.550	5.271	1.500
45	6.978	11.212	8.378	1.500	77.604	5.270	1.500
46	7.180	10.343	8.341	1.500	46.669	6.014	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 41

Governing Disturbing Force : -18.896kN

Governing Restoring Force : 108.957kN

Minimum Sliding Ratio for the Critical Load Case : 5.766

Critical Load Case for Overturning about X-Direction : 21

Governing Overturning Moment : 27.878kNm

Governing Resisting Moment : 137.041kNm

Minimum Overturning Ratio for the Critical Load Case : 4.916

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 21

Governing Disturbing Force : 23.847kN

Governing Restoring Force : 105.772kN

Minimum Sliding Ratio for the Critical Load Case : 4.435

Critical Load Case for Overturning about Z-Direction : 41

Governing Overturning Moment : 31.799kNm

Governing Resisting Moment : 142.025kNm

Minimum Overturning Ratio for the Critical Load Case : 4.466

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 28.372kN

Governing Restoring Force : 151.508kN

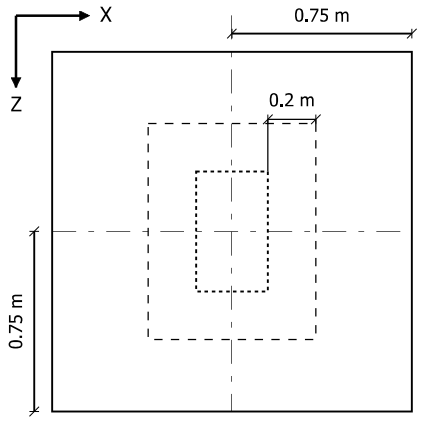
Minimum Sliding Ratio for the Critical Load Case : 5.340

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.667

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 278.154kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.976m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta_c}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2212.777kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3331.062kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2011.615kN

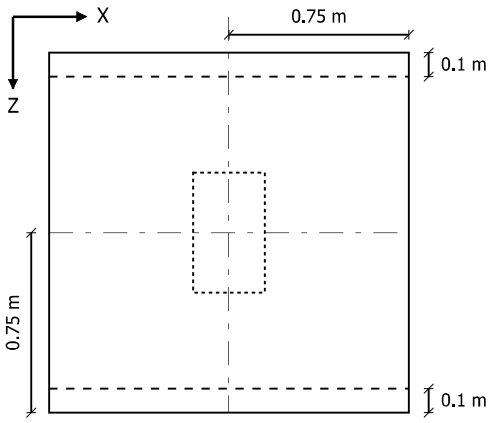
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1508.712kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.156m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

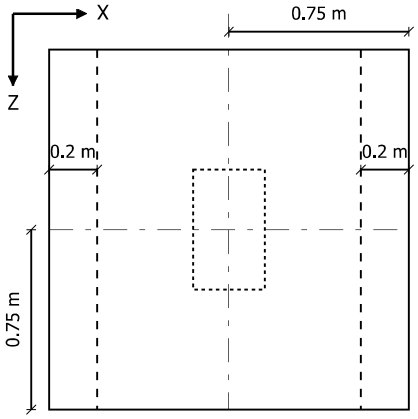
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{ux} is # 32 $V_{ux} = V_{ux}|_{x=D_x}$ = 49.518 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.256 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

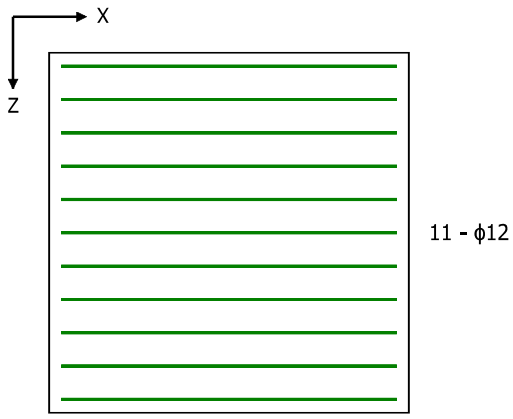
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z}$ = 87.559 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00173
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.600 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	58.964 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	65.515 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00088
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00076
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1038.302 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.550	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	11	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1244.062	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00241	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

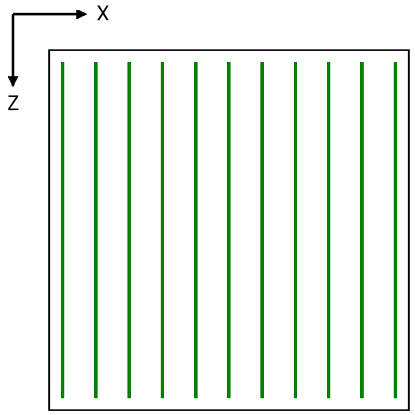
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.198

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



11 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 32

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	

From ACI Cl.7.12.2, ρ_{min} = 0.00171

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.000 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 37.922 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 42.136 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00061

(Based on gross depth) ρ x d_{eff} / Depth = 0.00051

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1023.991 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.450 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 11

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1244.062 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

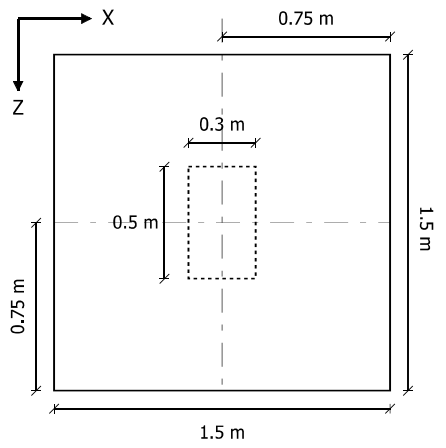
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00250

From ACI Cl.7.6.1, minimum req'd clear distance between bars

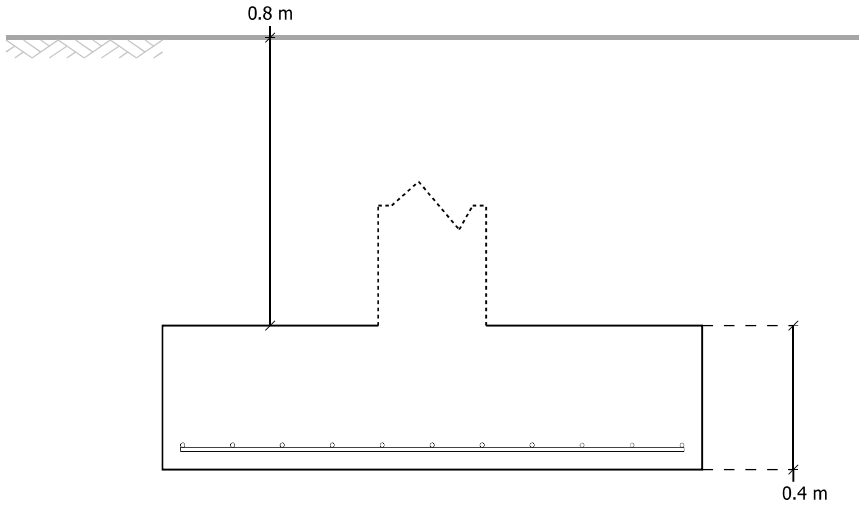
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.215

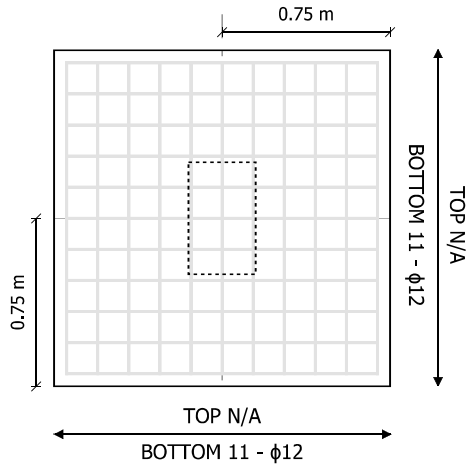
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

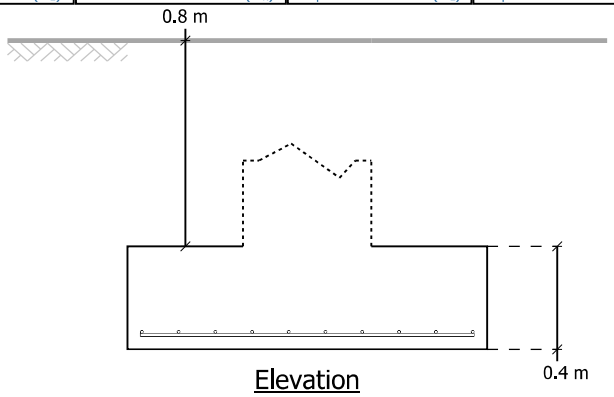
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.5 m	1.5 m	0.4 m	N/A	N/A	11 - ϕ12	11 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

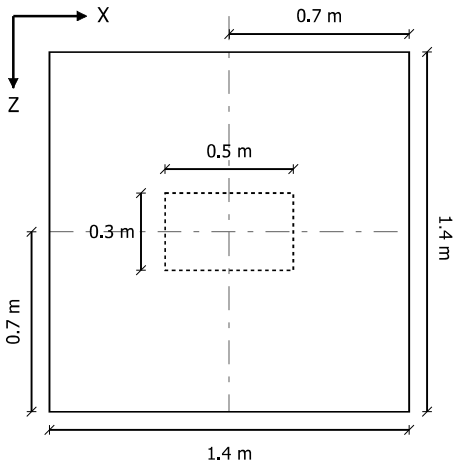
FOUNDATION-F3-GRID 6 & G

Footing No.	Group ID	Foundation Geometry		
		Length	Width	Thickness
-	-			

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
-						



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1400.000mm
Footing Width - Z (Fw) : 1400.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m
Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	24.000kN/m3
Strength of Concrete :	35.000N/mm2
Yield Strength of Steel :	420.000N/mm2
Minimum Bar Size :	#12
Maximum Bar Size :	#12
Top Footing Minimum Bar Size :	#12
Top Footing Maximum Bar Size :	#12
Pedestal Minimum Bar Size :	#16
Pedestal Maximum Bar Size :	#16
Minimum Bar Spacing :	100.000mm
Maximum Bar Spacing :	150.000mm
Pedestal Clear Cover (P, CL) :	50.000mm
Bottom Footing Clear Cover (F, CL) :	50.000mm

Soil Properties

Soil Type :	Cohesionless Soil
Unit Weight :	18.000kN/m3
Soil Bearing Capacity :	570.000kN/m2
Soil Bearing Capacity Type:	Net Bearing Capacity
Soil Surcharge :	0.000kN/m2
Depth of Soil above Footing :	800.000mm
Type of Depth :	Fixed Top
Undrained Shear Strength :	0.000kN/m2
Bearing Capacity Input Method:	Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction :	0.580
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Global Settings

Top Reinforcement Option :	Calculate only when foundation is subjected to uplift forces
Concrete Design Option :	Gross Pressure
Top Reinforcement Factor :	1.000

Design Calculations

Footing Size

Initial Length (L_o) =	1.400m
Initial Width (W_o) =	1.400m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1,2D+1,6WZ1+0,5LR)	1.00	1.00	1.00
29	ULS: (1,2D+1,6WX1+0,5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	362.551	5.808	-0.311	2.065	-1.730
12	375.426	6.685	-0.352	2.307	-2.144
13	372.208	6.466	-0.342	2.246	-2.041
14	362.545	5.898	7.786	26.897	-1.835
15	361.969	14.487	-0.479	1.610	-12.403
16	372.002	9.607	17.798	41.287	-5.447
17	372.084	8.091	15.256	35.845	-3.446
18	217.256	7.673	24.000	53.292	-5.580
19	217.366	5.652	20.611	46.036	-2.911
20	216.949	12.164	-0.355	0.784	-11.711
21	217.525	3.575	7.911	26.072	-1.143
22	362.276	9.996	23.876	54.118	-6.272
23	362.386	7.975	20.487	46.862	-3.603

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
39	372.434	3.206	17.904	41.567	1.466
40	372.098	7.743	-19.902	-39.885	-3.121
41	217.832	-0.862	24.142	53.667	3.638
42	217.384	5.188	-26.266	-54.937	-2.479
45	362.852	1.461	24.017	54.492	2.946
46	362.404	7.511	-26.391	-54.111	-3.171

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	507.571	8.131	-0.435	2.891	-2.422
25	514.009	8.569	-0.456	3.012	-2.629
26	514.009	8.569	-0.456	3.012	-2.629
27	455.442	11.724	18.910	44.508	-6.373
28	441.235	10.876	32.883	74.275	-5.280
29	441.059	14.109	38.305	85.884	-9.551
30	441.235	10.876	32.883	74.275	-5.280
31	434.230	19.369	-0.613	1.828	-17.324
32	435.053	7.099	11.194	37.953	-2.225
33	325.856	11.928	38.419	85.144	-8.825
34	326.032	8.695	32.996	73.534	-4.554
35	326.288	5.357	11.287	37.334	-1.706
36	325.465	17.627	-0.520	1.209	-16.805
47	455.903	4.896	19.024	44.808	1.001
48	441.264	10.133	-42.121	-87.282	-4.588
49	441.981	0.454	38.532	86.483	5.198
50	441.264	10.133	-42.121	-87.282	-4.588
53	326.778	-1.728	38.646	85.743	5.924
54	326.061	7.952	-42.007	-88.023	-3.862

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.960 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.224 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.400 m Governing Load Case : # 11

Width (W_2) = 1.400 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 50

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.960 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 18.815 kN

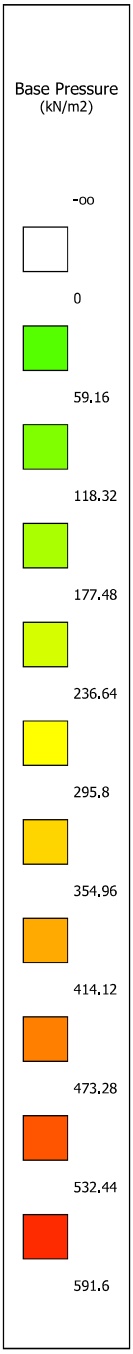
Gross Soil Bearing Capacity = 591.60 kN/m2

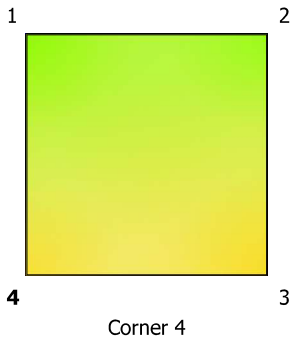
Soil Weight On Top Of Footing = 26.063 kN

Pressures at 4 Corners

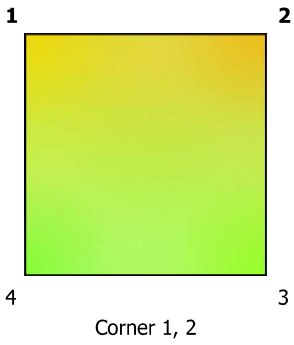
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

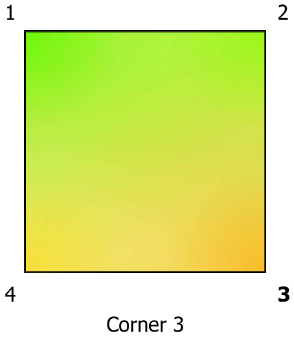




Corner 4

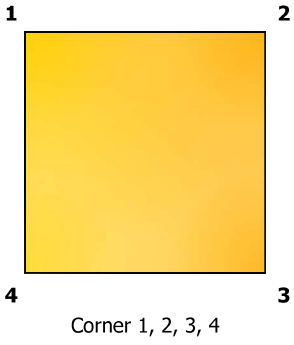


Corner 1, 2



Corner 3

Governing Base Pressure Contour



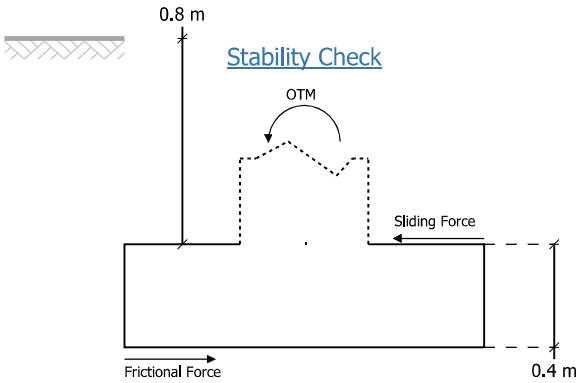
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**335.6978**	362.7026	79.8968	52.8920	0.000
46	335.6978	**362.7026**	79.8968	52.8920	0.000
22	46.0556	90.9706	**369.4082**	324.4931	0.000
45	73.0281	62.7008	343.0235	**353.3508**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**335.6978**	362.7026	79.8968	52.8920
46	335.6978	**362.7026**	79.8968	52.8920
22	46.0556	90.9706	**369.4082**	324.4931
45	73.0281	62.7008	343.0235	**353.3508**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	40.690	759.978	57.462	1.500	147.527	70.432	1.500
12	36.466	692.031	51.499	1.500	136.311	61.111	1.500
13	37.414	707.477	52.838	1.500	138.890	63.152	1.500
14	40.063	30.350	34.212	1.500	9.538	68.061	1.500
15	16.288	492.638	23.022	1.500	201.524	15.663	1.500
16	25.168	13.585	16.907	1.500	6.050	31.438	1.500
17	29.888	15.852	19.805	1.500	6.983	43.716	1.500
18	19.815	6.335	8.533	1.500	2.934	21.244	1.500
19	26.910	7.380	10.065	1.500	3.401	35.541	1.500
20	12.484	428.234	17.647	1.500	286.997	11.071	1.500
21	42.567	19.239	24.794	1.500	6.319	71.491	1.500
22	23.625	9.891	12.903	1.500	4.493	27.775	1.500
23	29.619	11.530	15.195	1.500	5.197	42.003	1.500
39	75.499	13.519	18.819	1.500	6.016	1591.484	1.500
40	31.233	12.152	16.016	1.500	6.122	46.976	1.500
41	176.779	6.312	8.920	1.500	2.921	46.237	1.500
42	29.320	5.791	8.035	1.500	2.821	40.369	1.500
45	161.856	9.846	13.899	1.500	4.469	120.966	1.500
46	31.450	8.951	12.175	1.500	4.425	46.210	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20

Governing Disturbing Force : 12.164kN

Governing Restoring Force : 151.860kN

Minimum Sliding Ratio for the Critical Load Case : 12.484

Critical Load Case for Overturning about X-Direction : 42

Governing Overturning Moment : -65.443kNm

Governing Resisting Moment : 184.639kNm

Minimum Overturning Ratio for the Critical Load Case : 2.821

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 42

Governing Disturbing Force : -26.266kN

Governing Restoring Force : 152.112kN

Minimum Sliding Ratio for the Critical Load Case : 5.791

Critical Load Case for Overturning about Z-Direction : 20

Governing Overturning Moment : -16.577kNm

Governing Resisting Moment : 183.528kNm

Minimum Overturning Ratio for the Critical Load Case : 11.071

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 42

Governing Disturbing Force : 26.774kN

Governing Restoring Force : 215.119kN

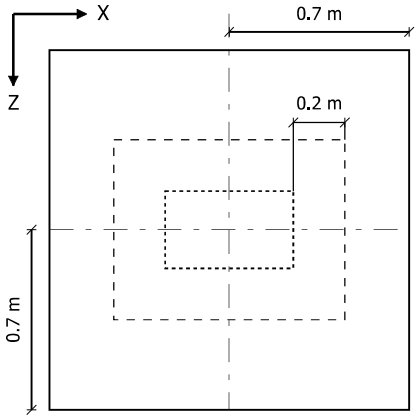
Minimum Sliding Ratio for the Critical Load Case : 8.035

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.667

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 403.900kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.976m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2212.777kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3331.062kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 2011.615kN

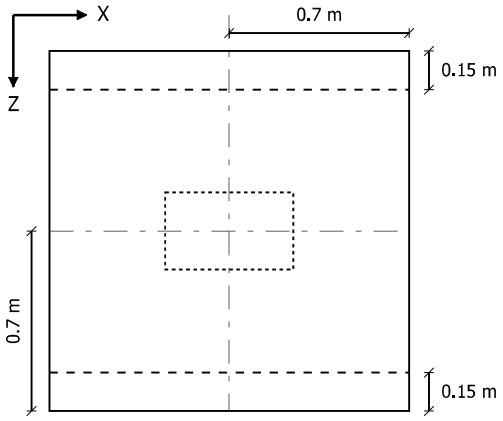
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1508.712kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 473.162kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.206m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

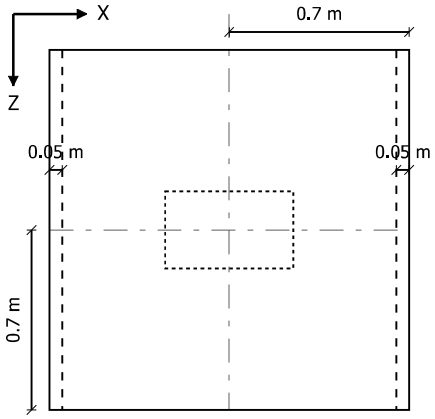
From above calculations, 0.75 X V_c = 354.872 kN

Critical load case for V_{ux} is # 50 $V_{ux} = V_{ux}|_{x=D_x}$ = 127.537 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 473.162 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.106 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

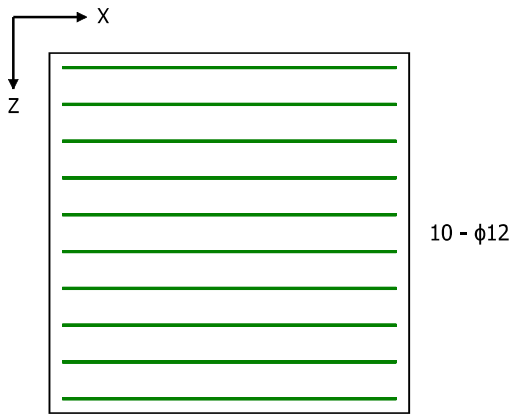
From above calculations, 0.75 X V_c = 354.872 kN

Critical load case for V_{uz} is # 25 $V_{uz} = V_{uz}|_{z=D_z}$ = 44.132 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00172
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.950 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	41.897 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	46.552 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00067
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00058
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	960.904 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.400	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	10	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1130.965	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00235	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

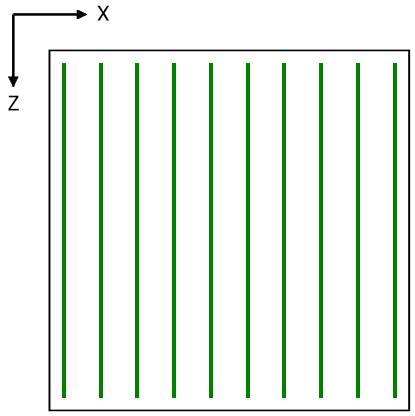
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.177

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



10 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 48

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	

From ACI Cl.7.12.2, ρ_{min} = 0.00175

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.550 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 88.096 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 97.884 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00153

(Based on gross depth) ρ x d_{eff} / Depth = 0.00127

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 978.605 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.500 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 10

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1130.965 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

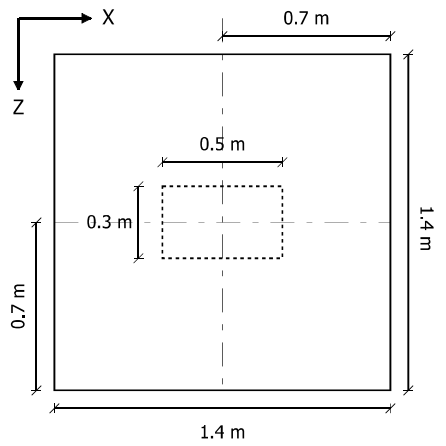
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00243

From ACI Cl.7.6.1, minimum req'd clear distance between bars

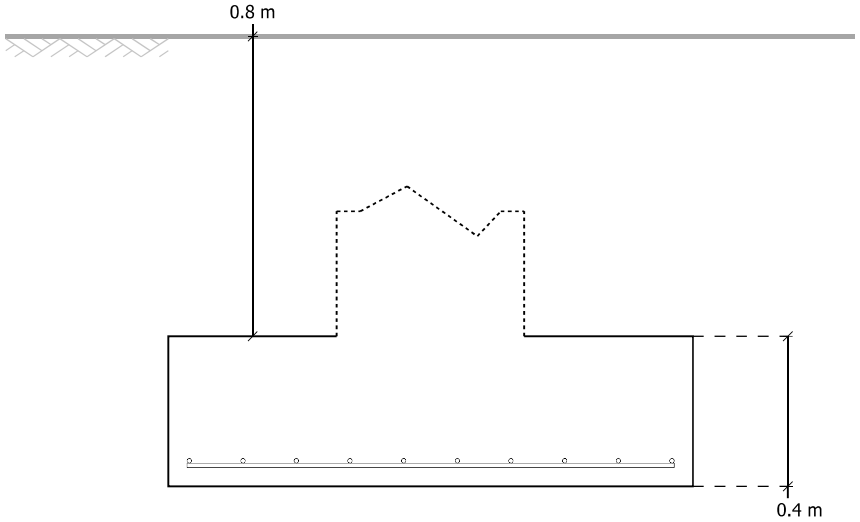
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.156

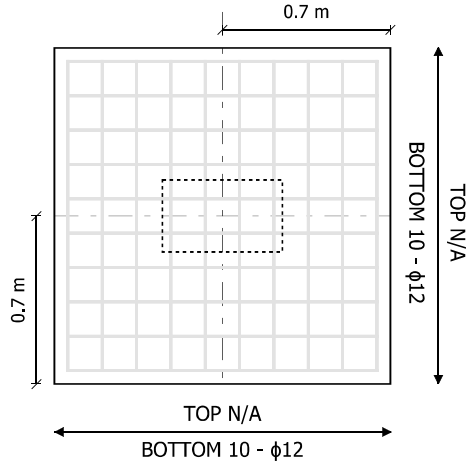
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

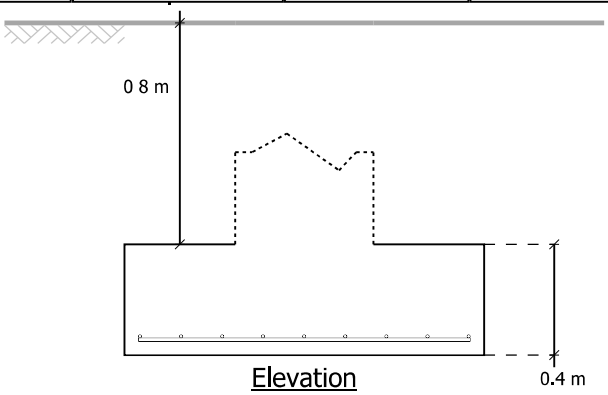
	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	1.4 m	1.4 m	0.4 m	N/A	N/A	10 - ϕ12	10 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

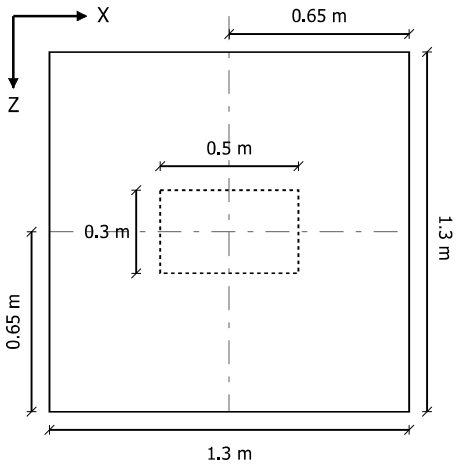
FOUNDATION-F4-GRID 1 & F

Footing No.	Group ID	Foundation Geometry		
		Length	Width	Thickness
-	-			

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
-						



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1300.000mm
Footing Width - Z (Fw) : 1300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m
Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	24.000kN/m3
Strength of Concrete :	35.000N/mm2
Yield Strength of Steel :	420.000N/mm2
Minimum Bar Size :	#12
Maximum Bar Size :	#12
Top Footing Minimum Bar Size :	#12
Top Footing Maximum Bar Size :	#12
Pedestal Minimum Bar Size :	#16
Pedestal Maximum Bar Size :	#16
Minimum Bar Spacing :	100.000mm
Maximum Bar Spacing :	150.000mm
Pedestal Clear Cover (P, CL) :	50.000mm
Bottom Footing Clear Cover (F, CL) :	50.000mm

Soil Properties

Soil Type :	Cohesionless Soil
Unit Weight :	18.000kN/m3
Soil Bearing Capacity :	570.000kN/m2
Soil Bearing Capacity Type:	Net Bearing Capacity
Soil Surcharge :	0.000kN/m2
Depth of Soil above Footing :	800.000mm
Type of Depth :	Fixed Top
Undrained Shear Strength :	0.000kN/m2
Bearing Capacity Input Method:	Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction :	0.580
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Global Settings

Top Reinforcement Option :	Calculate only when foundation is subjected to uplift forces
Concrete Design Option :	Gross Pressure
Top Reinforcement Factor :	1.000

Design Calculations

Footing Size

Initial Length (L_o) =	1.300m
Initial Width (W_o) =	1.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	339.591	0.907	-2.822	-3.061	-1.160
12	350.501	0.945	-2.724	-3.129	-1.245
13	347.774	0.936	-2.748	-3.112	-1.224
14	321.110	0.775	-0.833	2.591	-1.288
15	339.661	9.710	-2.374	-2.738	-11.545
16	377.343	5.067	-9.649	-14.172	-5.163
17	315.950	1.182	4.885	8.928	-1.921
18	243.180	6.052	-10.894	-16.583	-5.948
19	161.323	0.872	8.484	14.217	-1.625
20	203.825	9.348	-1.245	-1.513	-11.081
21	185.273	0.412	0.296	3.815	-0.825
22	379.017	6.415	-12.023	-17.808	-6.412
23	297.159	1.235	7.356	12.992	-2.089

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
39	377.326	-2.733	-10.033	-14.448	3.297
40	373.023	1.784	-8.809	-12.697	-1.481
41	243.158	-4.348	-11.406	-16.951	5.332
42	237.420	1.676	-9.775	-14.616	-1.039
45	378.994	-3.985	-12.535	-18.176	4.868
46	373.256	2.038	-10.904	-15.841	-1.503

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	475.427	1.270	-3.951	-4.285	-1.624
25	480.882	1.289	-3.902	-4.319	-1.666
26	480.882	1.289	-3.902	-4.319	-1.666
27	456.507	5.556	-10.590	-15.580	-5.730
28	345.074	1.632	12.947	21.978	-2.921
29	476.046	9.921	-18.059	-27.302	-9.838
30	345.074	1.632	12.947	21.978	-2.921
31	407.609	13.665	-2.747	-3.212	-16.227
32	381.107	0.900	-0.544	4.401	-1.576
33	368.713	9.629	-17.261	-26.350	-9.447
34	237.741	1.341	13.745	22.930	-2.531
35	279.230	0.628	0.302	5.319	-1.228
36	305.732	13.393	-1.900	-2.293	-15.880
47	456.489	-2.764	-11.000	-15.874	3.294
48	466.829	2.918	-16.268	-24.155	-1.984
49	476.010	-6.720	-18.878	-27.891	8.210
50	466.829	2.918	-16.268	-24.155	-1.984
53	368.677	-7.011	-18.081	-26.939	8.600
54	359.496	2.626	-15.470	-23.203	-1.593

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.690 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.952 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.300 m Governing Load Case : # 11

Width (W_2) = 1.300 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.690 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 16.223 kN

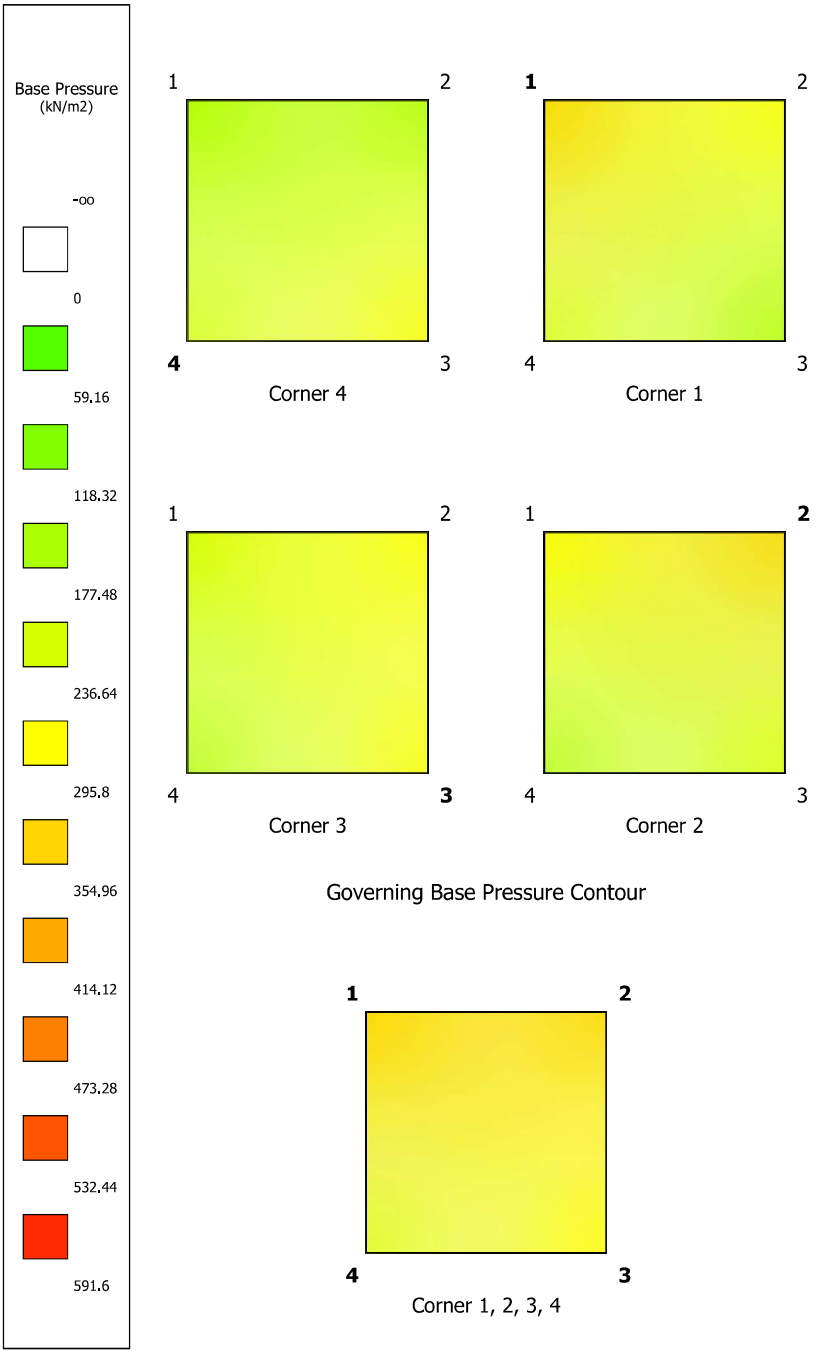
Gross Soil Bearing Capacity = 591.60 kN/m2

Soil Weight On Top Of Footing = 22.175 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

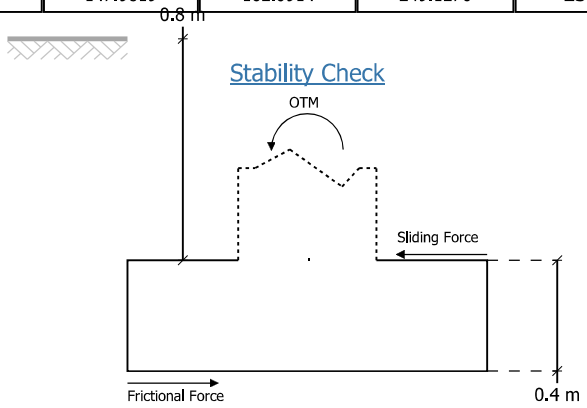


Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**327.9578**	292.6628	165.9982	201.2933	0.000
22	284.2398	**333.2777**	209.7428	160.7050	0.000
15	191.6377	275.9116	**255.7699**	171.4961	0.000
23	147.9819	162.0914	249.1276	**235.0182**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**327.9578**	292.6628	165.9982	201.2933
22	284.2398	**333.2777**	209.7428	160.7050
15	191.6377	275.9116	**255.7699**	171.4961
23	147.9819	162.0914	249.1276	**235.0182**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	241.684	77.685	104.593	1.500	58.877	161.523	1.500
12	238.601	82.815	110.643	1.500	60.155	155.890	1.500
13	239.348	81.498	109.105	1.500	59.837	157.232	1.500
14	268.965	250.435	259.205	1.500	103.924	146.330	1.500
15	22.581	92.353	31.021	1.500	66.906	15.942	1.500
16	47.589	24.991	31.290	1.500	15.041	37.619	1.500
17	173.933	42.072	57.831	1.500	21.256	96.331	1.500
18	26.984	14.991	18.533	1.500	8.787	21.898	1.500
19	132.838	13.653	19.207	1.500	7.427	65.877	1.500
20	15.029	112.800	21.069	1.500	78.753	10.640	1.500
21	314.568	437.939	361.317	1.500	37.207	147.163	1.500
22	37.738	20.137	25.125	1.500	12.040	30.247	1.500
23	157.606	26.459	36.902	1.500	13.749	84.525	1.500
39	88.215	24.032	32.792	1.500	14.690	61.601	1.500
40	133.738	27.087	37.545	1.500	16.547	121.949	1.500
41	37.559	14.317	18.919	1.500	8.552	25.915	1.500
42	95.475	16.366	22.812	1.500	9.730	105.017	1.500
45	60.749	19.313	26.028	1.500	11.742	42.021	1.500
46	117.130	21.897	30.440	1.500	13.293	115.514	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20

Governing Disturbing Force : 9.348kN

Governing Restoring Force : 140.489kN

Minimum Sliding Ratio for the Critical Load Case : 15.029

Critical Load Case for Overturning about X-Direction : 19

Governing Overturning Moment : 17.610kNm

Governing Resisting Moment : 130.799kNm

Minimum Overturning Ratio for the Critical Load Case : 7.427

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19

Governing Disturbing Force : 8.484kN

Governing Restoring Force : 115.838kN

Minimum Sliding Ratio for the Critical Load Case : 13.653

Critical Load Case for Overturning about Z-Direction : 20

Governing Overturning Moment : -14.820kNm

Governing Resisting Moment : 157.676kNm

Minimum Overturning Ratio for the Critical Load Case : 10.640

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 18

Governing Disturbing Force : 12.462kN

Governing Restoring Force : 230.964kN

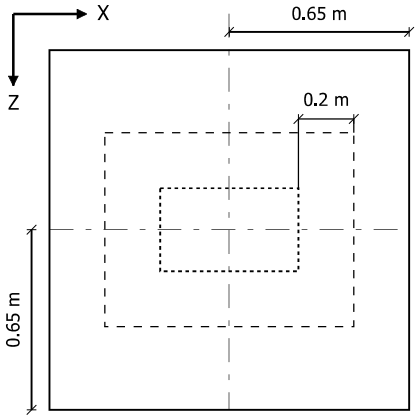
Minimum Sliding Ratio for the Critical Load Case : 18.533

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.667

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 352.270kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.976m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2212.777kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3331.062kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2011.615kN

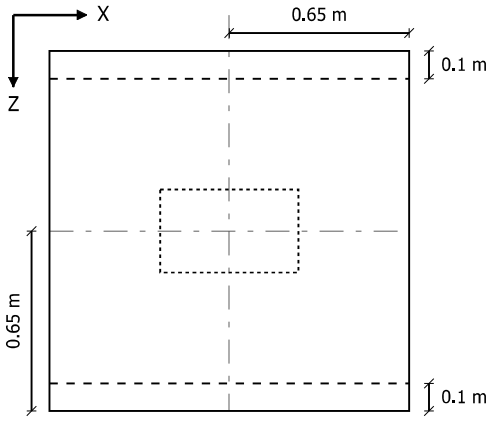
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1508.712kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.156m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

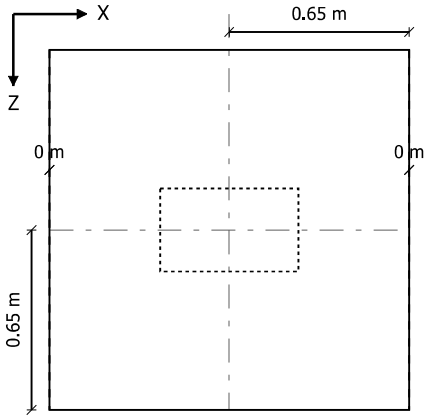
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{ux} is # 49 $V_{ux} = V_{ux}|_{x=D_x}$ = 79.003 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.056 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

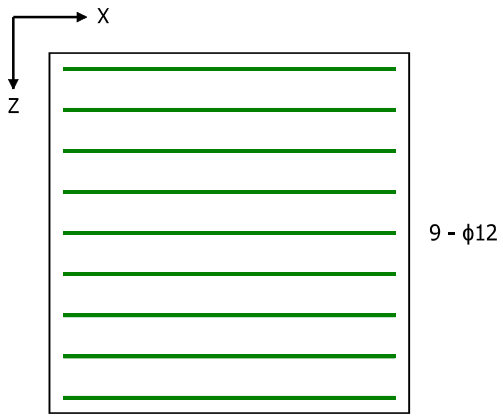
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{uz} is # 29 $V_{uz} = V_{uz}|_{z=D_z} =$ 24.787 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00173
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.900 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	34.782 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	38.646 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00060
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00052
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	899.862 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.350	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	9	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1017.869	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00228	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

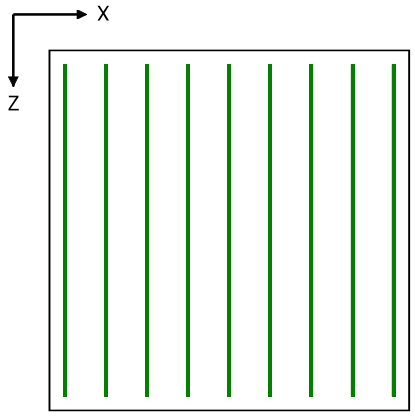
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.131

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	

From ACI Cl.7.12.2, ρ_{min} = 0.00176

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.500 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 61.154 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 67.949 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00114

(Based on gross depth) ρ x d_{eff} / Depth = 0.00094

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 914.436 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 148.500mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.450 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

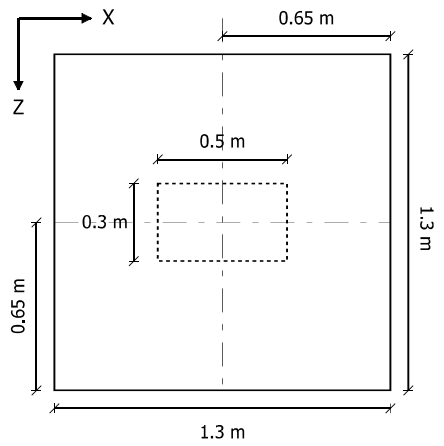
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00236

From ACI Cl.7.6.1, minimum req'd clear distance between bars

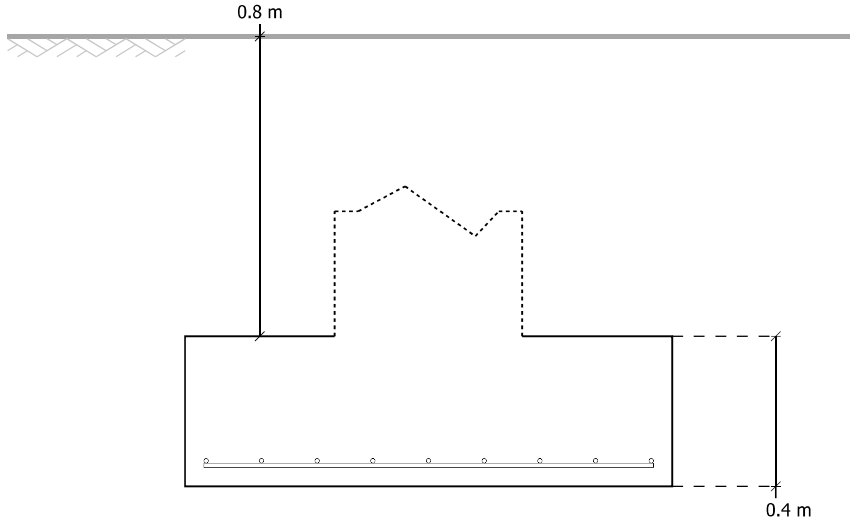
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.113

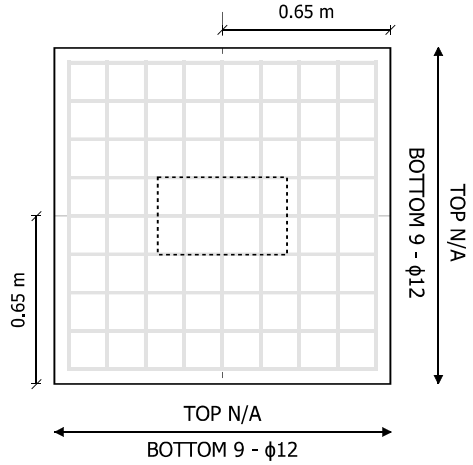
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

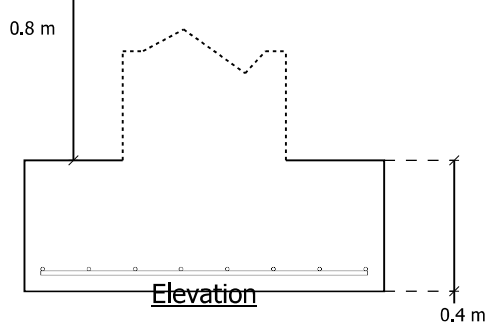
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.3 m	1.3 m	0.4 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

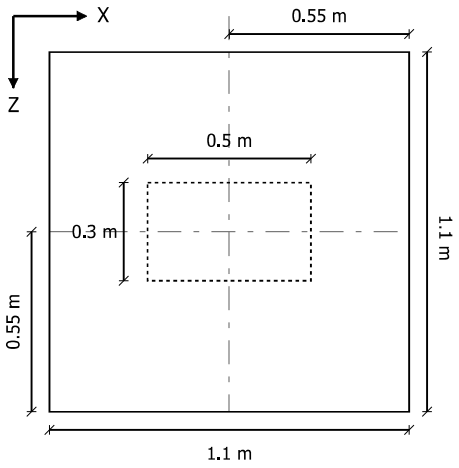
FOUNDATION-F5-GRID 4 & B

Footing No.	Group ID	Foundation Geometry		
		Length	Width	Thickness
-	-			

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
-						



Elevation



Plan

Input Values

Footing Geomtery

Design Type :	Set Dimension
Footing Thickness (Ft) :	400.000mm
Footing Length - X (Fl) :	1100.000mm
Footing Width - Z (Fw) :	1100.000mm
Eccentricity along X (Oxd) :	0.000mm
Eccentricity along Z (Ozd) :	0.000mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (D_{col}) :	0.500m
Column Width - Z (B_{col}) :	0.300m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	24.000kN/m3
Strength of Concrete :	35.000N/mm2
Yield Strength of Steel :	420.000N/mm2
Minimum Bar Size :	#12
Maximum Bar Size :	#12
Top Footing Minimum Bar Size :	#12
Top Footing Maximum Bar Size :	#12
Pedestal Minimum Bar Size :	#16
Pedestal Maximum Bar Size :	#16
Minimum Bar Spacing :	100.000mm
Maximum Bar Spacing :	150.000mm
Pedestal Clear Cover (P, CL) :	50.000mm
Bottom Footing Clear Cover (F, CL) :	50.000mm

Soil Properties

Soil Type :	Cohesionless Soil
Unit Weight :	18.000kN/m3
Soil Bearing Capacity :	570.000kN/m2
Soil Bearing Capacity Type:	Net Bearing Capacity
Soil Surcharge :	0.000kN/m2
Depth of Soil above Footing :	800.000mm
Type of Depth :	Fixed Top
Undrained Shear Strength :	0.000kN/m2
Bearing Capacity Input Method:	Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction :	0.580
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Global Settings

Top Reinforcement Option :	Calculate only when foundation is subjected to uplift forces
Concrete Design Option :	Gross Pressure
Top Reinforcement Factor :	1.000

Design Calculations

Footing Size

Initial Length (L_o) =	1.100m
Initial Width (W_o) =	1.100m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1,2D+1,6WZ1+0,5LR)	1.00	1.00	1.00
29	ULS: (1,2D+1,6WX1+0,5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	244.515	-1.763	0.501	-0.714	-0.366
12	250.876	-2.132	0.558	-0.820	-0.198
13	249.286	-2.040	0.544	-0.793	-0.240
14	244.514	-1.641	4.800	10.696	-0.528
15	245.667	6.222	0.052	-1.739	-9.199
16	249.748	1.937	-12.691	-23.980	-4.236
17	249.202	-3.001	15.134	25.294	0.383
18	147.326	4.244	-17.346	-31.344	-5.547
19	146.598	-2.340	19.754	34.355	0.611
20	147.860	6.927	-0.148	-1.454	-9.053
21	146.708	-0.936	4.599	10.981	-0.382
22	245.132	3.539	-17.145	-31.629	-5.694
23	244.404	-3.045	19.954	34.070	0.465

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
39	248.717	-6.350	-12.405	-23.371	4.300
40	249.166	-3.549	-11.131	-21.800	1.048
41	145.951	-6.805	-16.964	-30.532	5.833
42	146.549	-3.070	-15.266	-28.436	1.498
45	243.757	-7.510	-16.763	-30.817	5.687
46	244.355	-3.776	-15.065	-28.722	1.352

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	342.322	-2.468	0.702	-0.999	-0.513
25	345.502	-2.653	0.730	-1.052	-0.429
26	345.502	-2.653	0.730	-1.052	-0.429
27	304.089	1.536	-13.425	-25.759	-4.432
28	296.421	-4.351	31.755	54.744	0.974
29	297.585	6.183	-27.605	-50.375	-8.879
30	296.421	-4.351	31.755	54.744	0.974
31	295.063	9.292	-0.040	-2.321	-13.058
32	293.416	-1.941	6.742	15.443	-0.671
33	221.050	6.896	-27.783	-50.107	-8.853
34	219.886	-3.638	31.576	55.012	1.000
35	220.062	-1.412	6.592	15.657	-0.561
36	221.708	9.821	-0.190	-2.107	-12.948
47	302.989	-7.303	-13.119	-25.109	4.672
48	296.343	-5.520	-24.276	-45.723	2.394
49	295.385	-11.495	-26.993	-49.075	9.329
50	296.343	-5.520	-24.276	-45.723	2.394
53	218.850	-10.782	-27.172	-48.808	9.355
54	219.808	-4.807	-24.455	-45.455	2.419

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.210m^2

Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 0.884m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.100 m Governing Load Case : # 11

Width (W_2) = 1.100 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 30

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.210 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 11.616 kN

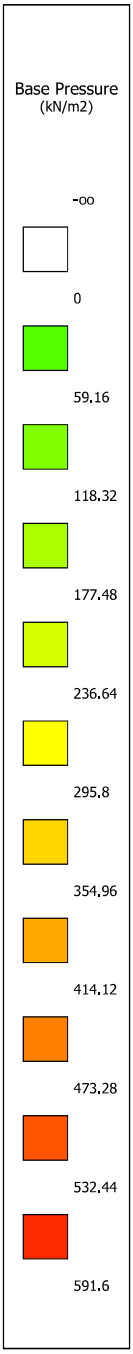
Gross Soil Bearing Capacity = 591.60 kN/m2

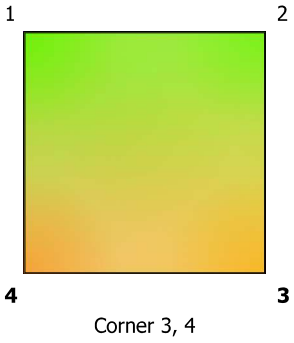
Soil Weight On Top Of Footing = 15.263 kN

Pressures at 4 Corners

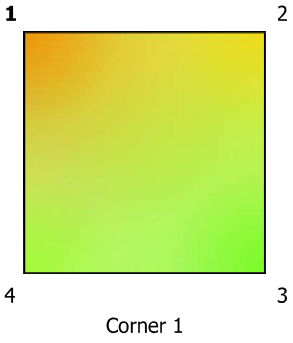
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

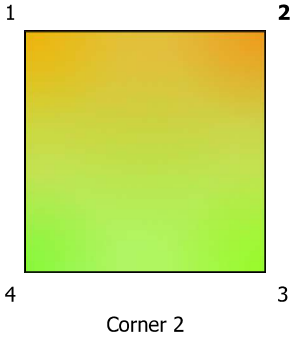




Corner 3, 4

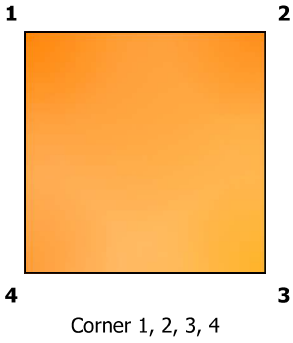


Corner 1



Corner 2

Governing Base Pressure Contour



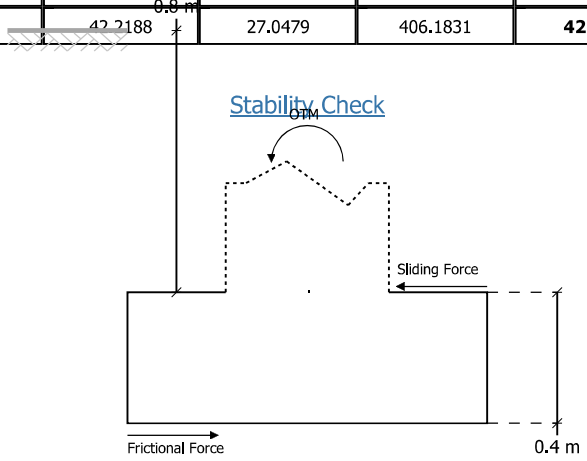
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**431.9937**	353.6380	15.3385	93.6942	0.000
22	366.2550	**430.3490**	83.3499	19.2559	0.000
23	42.2188	27.0479	**406.1831**	421.3540	0.000
23	42.2188	27.0479	406.1831	**421.3540**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**431.9937**	353.6380	15.3385	93.6942
22	366.2550	**430.3490**	83.3499	19.2559
23	42.2188	27.0479	**406.1831**	421.3540
23	42.2188	27.0479	406.1831	**421.3540**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	89.275	314.142	121.445	1.500	292.455	440.853	1.500
12	75.574	288.673	103.393	1.500	257.352	233.695	1.500
13	78.535	294.540	107.316	1.500	265.172	264.196	1.500
14	95.930	32.795	43.886	1.500	11.897	1168.959	1.500
15	25.405	3029.213	35.927	1.500	87.717	12.842	1.500
16	82.837	12.642	17.674	1.500	5.265	30.405	1.500
17	53.361	10.581	14.678	1.500	4.870	96.022	1.500
18	23.808	5.825	8.002	1.500	2.524	13.253	1.500
19	43.005	5.093	7.153	1.500	2.278	61.800	1.500
20	14.630	683.654	20.685	1.500	64.067	8.145	1.500
21	107.611	21.891	30.337	1.500	7.511	12338.939	1.500
22	44.583	9.202	12.745	1.500	3.909	21.072	1.500
23	51.674	7.885	11.024	1.500	3.568	88.787	1.500
39	25.173	12.886	16.222	1.500	5.379	22.191	1.500
40	45.114	14.384	19.381	1.500	5.815	61.596	1.500
41	14.730	5.909	7.756	1.500	2.570	11.134	1.500
42	32.760	6.589	9.136	1.500	2.785	35.056	1.500
45	20.900	9.364	12.085	1.500	3.989	17.150	1.500
46	41.665	10.442	14.325	1.500	4.317	52.189	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20

Governing Disturbing Force : 6.927kN

Governing Restoring Force : 101.349kN

Minimum Sliding Ratio for the Critical Load Case : 14.630

Critical Load Case for Overturning about X-Direction : 19

Governing Overturning Moment : 42.257kNm

Governing Resisting Moment : 96.242kNm

Minimum Overturning Ratio for the Critical Load Case : 2.278

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19

Governing Disturbing Force : 19.754kN

Governing Restoring Force : 100.617kN

Minimum Sliding Ratio for the Critical Load Case : 5.093

Critical Load Case for Overturning about Z-Direction : 20

Governing Overturning Moment : -11.824kNm

Governing Resisting Moment : 96.303kNm

Minimum Overturning Ratio for the Critical Load Case : 8.145

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 19.892kN

Governing Restoring Force : 142.293kN

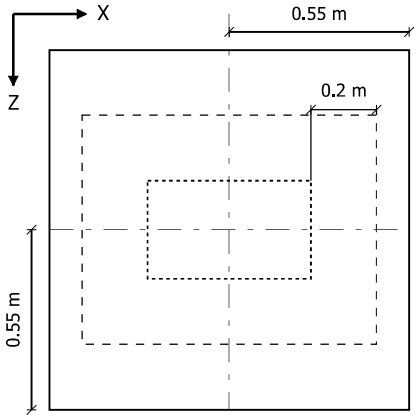
Minimum Sliding Ratio for the Critical Load Case : 7.153

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.667

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 205.106kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.976m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2212.777kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3331.062kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 2011.615kN

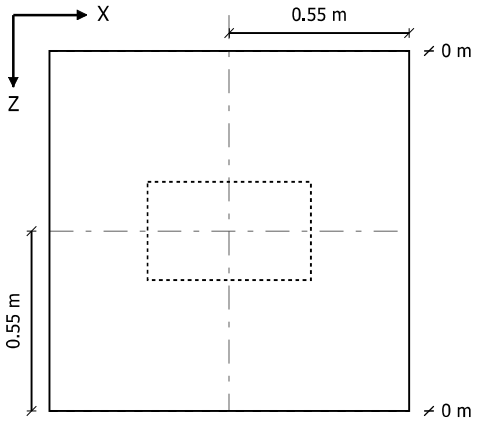
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1508.712kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 371.770kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.056m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

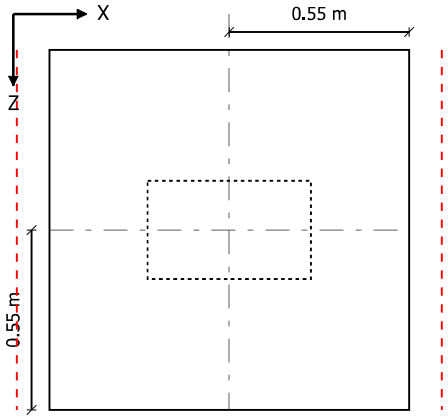
From above calculations, 0.75 X V_c = 278.828 kN

Critical load case for V_{ux} is # 30 $V_{ux} = V_{ux}|_{x=D_x}$ = 32.876 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 371.770 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.000 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

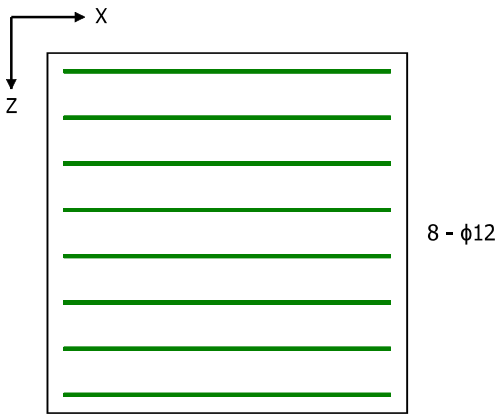
From above calculations, 0.75 X V_c = 278.828 kN

Critical load case for V_{uz} is # 24 $V_{uz} = V_{uz}|_{z=D_z}$ = 0.000 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 31

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00176
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.800 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	16.243 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	18.048 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00033
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00028
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	772.434 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.250	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	8	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	904.772	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00239	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

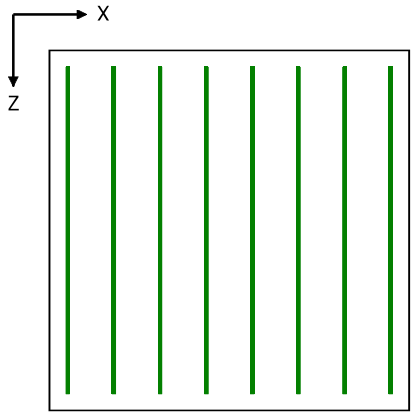
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.171

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



8 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 34

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	

From ACI Cl.7.12.2, ρ_{min} = 0.00178

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.700 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 44.776 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 49.751 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00098

(Based on gross depth) ρ x d_{eff} / Depth = 0.00082

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 781.738 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.350 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 8

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 904.772 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

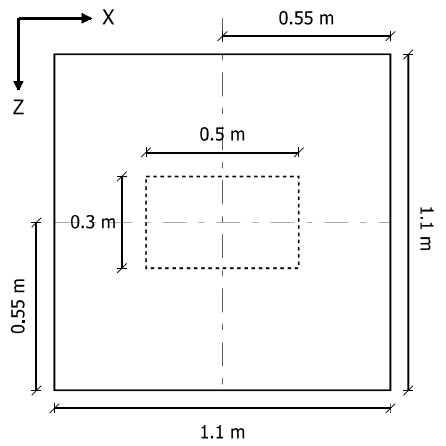
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00248

From ACI Cl.7.6.1, minimum req'd clear distance between bars

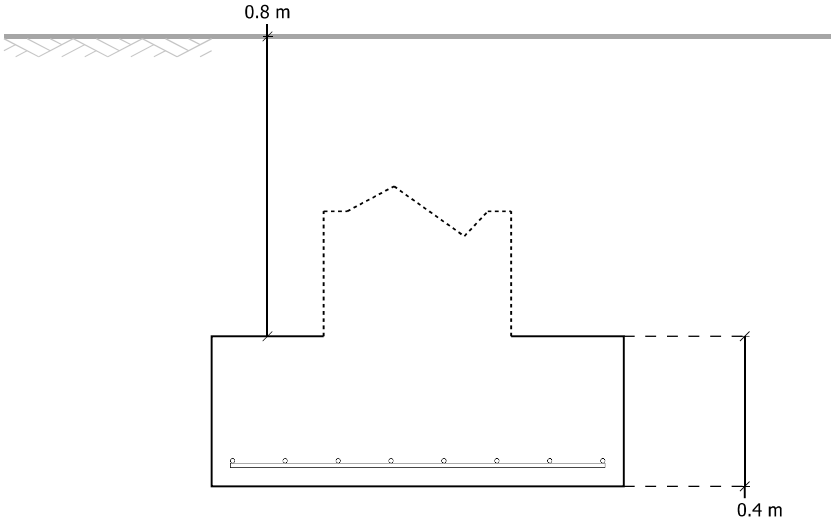
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.157

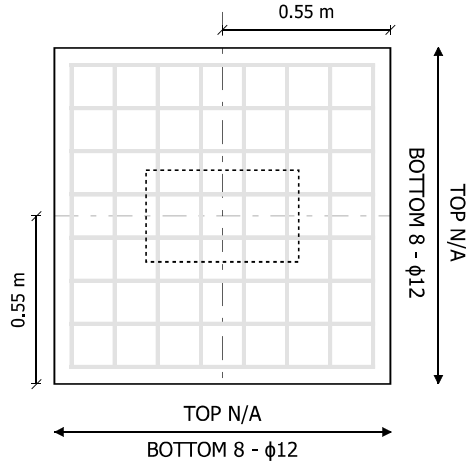
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	1.1 m	1.1 m	0.4 m	N/A	N/A	8 - ϕ12	8 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet